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RESEARCH ARTICLE

Green Container Shipping Schedule Design Considering Collaboration and Disruption Recovery

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ABSTRACT Emission reduction ambition in shipping and efforts to provide measures to achieve zero emission in the industry is increasing in parallel to global environmental concerns. Designing container shipping schedules that create minimum emissions can help to achieve the zero-emission target. This study proposed a comprehensive green container shipping scheduling model considering port and sea passage disruption recovery i.e., speed adjustment, port skipping, and collaboration of carrier and terminal operators because using only speed increase as a disruption recovery measure results in increased emissions. A mixed-integer multi-objective robust programming model was developed and applied to a real-case container line service demonstrating its benefits regarding emissions and operational costs. The results showed that different instances of the model can be solved in a reasonable time by a generic solver. The experimental results demonstrated that the most effective disruption recovery measure in terms of both operational costs and emission tonnages is the time window collaboration. The insights gathered from the application of the model can benefit container lines that are willing to reduce their emissions and can help policymakers provide effective emission reduction policies by taxing emissions and ship fuel.

INDEX TERMS Collaboration, container shipping, disruption recovery, emissions reduction, green schedule design.

I. INTRODUCTION

Container shipping is one of the submarkets of maritime transportation. Maritime transportation handles over 80% of international trade in goods and over 20% of that amount is carried by container shipping [1]. Container shipping is performed mostly by liner shipping where container lines operate their vessels on fixed routes and schedules. Container vessels perform round trips in fixed routes, which are called container line services, and the routes of each service and dates of vessel call to the ports of those services are declared to shippers. This fixed schedule provides certain advantages to the other players such as shippers (exporter and importers), terminal operators, and inland transportation service providers. Considering fixed schedules provided by container lines, container terminal operators plan internal operations [2] and their schedules for berth allocation, inland transport operators arrange their services, and shippers determine their manufacturing schedules, raw material procurement, and finished product distribution.

Additionally, vessel schedules must be designed by container lines in a way that minimizes total costs and/or maximizes total profits for them to compete in the market. Vessel schedule design in container shipping includes decisions such as vessel arrival time windows in ports, vessel speed, container handling rate, and the number of container vessels to allocate in the service [3]. Schedule design should also consider disruptions because they are inevitable. They might arise in sea passages or in ports. Disruptions in sea passages can be caused by bad weather, engine failures, and accidents, among others. On the other hand, disruptions in ports can be caused by labor strikes, natural disasters (earthquakes, storms, and hurricanes, etc.), labor shortages (like labor shortages due to COVID-19), and accidents in the arteries

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that connect ports with their hinterlands [4], [5]. Disruptions have consequences. Late arrivals of container vessels to the scheduled ports may not only result in penalties imposed by container terminal operators but also interrupt operations of inland transport operators and shippers causing financial loss [6]. In addition to financial loss, disruptions may increase the carbon footprint of container shipping because recovering from disruptions usually requires container vessels to speed up that in turn increases fuel consumption and emissions [7], [8], [9], [10], [11].

Emission reduction ambition in shipping and efforts to provide measures to achieve zero emission in the industry is increasing in parallel to global environmental concerns. Therefore, in 2018, IMO member states agreed to reduce the total annual GHG emissions resulting from ships by at least 50% by 2050 compared to 2008 [1]. Achieving this goal requires both new propulsion technologies that work with clean fuels such as hydrogen fuel cells, LNG, ammonia, etc., and operational measures such as designing container shipping schedules that produce as little emissions as possible [12]. In addition, IMOs emission reduction criteria can be achieved in a fast and effective way considering disruption recovery measures i.e., port skipping and carrier and terminal operator collaboration to design green container shipping schedules instead of speed increase because using only speed increase as a disruption recovery measure results in increased emissions. This study investigates the research question that how collaboration with terminal operators and port skipping can be used as disruption recovery measures in container shipping to reduce costs and environmental impacts of speed increase as a disruption recovery measure. In this regard, an optimization model was proposed to design robust container shipping schedules from an environmental perspective that accounts for disruptions and considers recovery measures i.e., speed adjustment, port skipping, and collaboration of carrier and terminal operators. For that purpose, a mixed-integer multi-objective robust programming model was developed. The model was applied to a real case container line service and computational experiments were performed to assess the benefits gained from the proposed green container shipping schedule design. The insights provided by the application of the model can benefit container lines that are willing to reduce their emissions and can help policymakers provide effective emission reduction policies by taxing emissions and ship fuel.

II. LITERATURE REVIEW

The provision of reliable vessel schedules can help container lines to increase their customer satisfaction and profits as well as environmental emissions. Therefore, much research has been conducted on vessel scheduling in container shipping [13], [14], [15], [16], [17], [18], [19] and container vessel speed optimization [20], [21], [22], [23], [24], [25]. For the sake of brevity, this section only presents the findings of the studies on container vessel scheduling that considers environmental scheduling, disruption recovery, uncertainty in port and sailing times, and collaboration with terminal operators as illustrated in Table 1.

TABLE 1. Related literature.

Reference	Uncertainty		Disruption Recovery		Collaboration		Modeling Approach		Objective Function		
	Port Time	Sail Time	Speed Adjustment	Port Skipping	Handling Rate	Time Window	Multi-Voyage	Multi-Ship	Emissions	Costs	
[8]					√	√			√	√	
[33]					√	√				√	
[15]					√	√			√	√	
[34]						√				√	
[35]					√					√	
[36]						√				√	
[7]						√				√	
[19]						√				√	
[27]			√	√						√	
[28]			√							√	
[31]			√	√						√	
[29]			√	√						√	
[30]			√	√	√					√	
[32]			√	√						√	
[37]	√	√	√							√	
[38]	√		√							√	
[10]	√		√				√		√	√	
[26]	√		√				√		√	√	
[39]	√		√	√						√	
[40]	√		√			√				√	
[41]	√		√					√		√	
[42]	√	√	√	√		√				√	
Current Study	√	√	√	√	√	√	√	√	√	√	

A. ENVIRONMENTAL CONTAINER VESSEL SCHEDULING

Maritime freight transportation is the most environmentally friendly transportation mode in terms of the amount of emissions produced as per ton kilometers of transportation activity, however, it produces high amount of emissions in total because the biggest portion of international freight is carried through maritime transportation [1]. Because container shipping is a submarket of maritime transportation, it applies to container shipping as well. Reducing the amount of emission in container shipping would require adoption of operational measures such as designing container shipping schedules that produce as little emissions as possible in addition to technological innovations such as new propulsion technologies that work with clean fuels such as hydrogen fuel cells, LNG, or ammonia [12]. Because bunker consumption highly depends on ship speed, designing liner shipping vessel schedules that require slow speed can reduce bunker consumption, thus, emissions. Qi and Song [10], Song et al. [26], Dulebenets [15], and Dulebenets [8] found that significant reduction in fuel consumption and emissions can be achieved by optimizing container vessel schedules. Qi and Song [10] and Song et al. [26] revealed that shorter legs of a container liner service remain to be problematic for emission reduction in cases where port disruptions occur, however, significant improvement in emission reduction is possible with sacrifices in service levels meaning arriving late to ports. Findings of Song et al. [26] suggested that to reduce carbon emission, the actual operation speed of ships planned by liner companies should be much lower than the maximum design speed of vessels. Additionally, Dulebenets [15] underlined that to reduce vessel speed and emissions, container liner companies need to allocate more vessels in their services. Dulebenets [8]

TABLE 2. Nomenclature.

Sets	
P	Set of port calls
Q	Set of ships
R	Set of port handling rates
L	Set of line segments for piecewise linear bunker consumption function
V	Set of voyages
Parameters	
c_r^{hand}	Cost of handing (per TEU) for handling rate $r \in R$
c_i^{late}	Cost of late arrival per hour at port $i \in P$
$c_{ij}^{road_div}$	Cost of container diversion between port $i \in P$ and $j \in P$
c^{fuel}	Cost of fuel per ton
$e_{ij}^{road_div}$	Emission factor as t of CO2 per TEU transported by road between port $i \in P$ and $j \in P$
e^{fuel}	Emission factor as t of CO2 per ton of fuel
e^{hand}	Emission factor as t of CO2 per TEU handled
c^{ems}	Cost of emissions per ton of CO2
c^{round}	The cost incurred from a round trip of diverted incoming containers
c^{sea_div}	The cost incurred from diverting constrainers through the sea
c_q^{voy}	Cost of operations per day for ship $q \in Q$
c_i^{stor}	Storage cost per week per TEU at port $i \in P$
d_{ijqv}	Demand for container transportation from port $i \in P$ to port $j \in P$ for ship $q \in Q$ on voyage $v \in V$
$dist_{ij}$	Distance of sea passage between port $i \in P$ and $j \in P$
$\delta_{iqv}^{port_occ}$	The binary parameter that equals 1 if a disruption occurs at port $i \in P$ to ship $q \in Q$ on voyage $v \in V$, otherwise equals 0
$\xi_{iqv}^{port_uur}$	Uncertain parameter for disruption duration at port $i \in P$ for ship $q \in Q$ on voyage $v \in V$, $\xi_{iqv}^{port_uur} = \bar{\xi}_{iqv}^{port_uur} + \hat{\xi}_{iqv}^{port_uur}$
$\bar{\xi}_{iqv}^{port_dur}$	Nominal value of disruption duration at port $i \in P$ for ship $q \in Q$ on voyage $v \in V$
$\hat{\xi}_{iqv}^{port_dur}$	Independent random parameter for disruption duration at port $i \in P$ for ship $q \in Q$ on voyage $v \in V$
$\delta_{iqv}^{sea_occ}$	The binary parameter that equals 1 if a disruption occurs at sea leg $i \in P$ to ship $q \in Q$ on voyage $v \in V$, otherwise equals 0
$\xi_{iqv}^{sea_dur}$	Uncertain parameter for disruption duration at sea leg $i \in P$ for ship $q \in Q$ on voyage $v \in V$, $\xi_{iqv}^{sea_dur} = \bar{\xi}_{iqv}^{sea_dur} + \hat{\xi}_{iqv}^{sea_dur}$
$\bar{\xi}_{iqv}^{sea_dur}$	Nominal value of disruption duration at sea leg $i \in P$ for ship $q \in Q$ on voyage $v \in V$
$\hat{\xi}_{iqv}^{sea_dur}$	Independent random parameter for disruption duration at sea leg $i \in P$ for ship $q \in Q$ on voyage $v \in V$
$\delta_{ij}^{str_pass}$	The binary parameter that equals 1 if sea passage from port $i \in P$ to port $j \in P$ has a strait passage, otherwise equals 0
$\delta_{ij}^{str_dur}$	Strait passage duration between port $i \in P$ and $j \in P$
$\delta_{ij}^{road_div}$	Availability of diversion between port $i \in P$ and $j \in P$
δ_i^{late}	The duration that must be waited in case of a late arrival to port $i \in P$
h_r^{rate}	Handling rate $r \in R$
γ	Bunker consumption coefficient for ship speed
α	Bunker consumption coefficient for ship speed
u_s^{min}	Minimum of the reciprocal of speed in line $s \in L$
u_s^{max}	Maximum of the reciprocal of speed in line $s \in L$
$slope_s$	The slope of line $s \in L$
$const_s$	Constant of line $s \in L$
f	Ship frequency
t_i^{fix}	Fixed time for pilotage and berthing operations in port $i \in P$
tw^{dur}	Time window duration
cap_q	The capacity of ship $q \in Q$
M^{late}	Maximum allowed late arrival hours for ships
Γ^{sea}	Scale parameter for uncertain disruption duration at sea
Γ^{port}	Scale parameter for uncertain disruption duration at port
M	Sufficiently big number
Decision Variables	
H_{irqv}^{rate}	The binary variable that equals 1 if handling rate $r \in R$ is chosen in port $i \in P$ for ship $q \in Q$ on voyage $v \in V$, otherwise equals 0
Tw_{iqv}^{st}	Staring time of time window at port $i \in P$ for ship $q \in Q$ on voyage $v \in V$
Tw_{iqv}^{end}	Ending time of time window at port $i \in P$ for ship $q \in Q$ on voyage $v \in V$
XY_{iiqv}	The binary variable that equals 1 if sea passage from port $i \in P$ to $j \in P$ for ship $q \in Q$ on voyage $v \in V$ is conducted, otherwise equals 0
X_{ijqv}	Number of TEUs transported from port $i \in P$ to $j \in P$ with ship $q \in Q$ on voyage $v \in V$
H_{iqv}	Number of TEUs loaded/unloaded at port $i \in P$ from/to ship $q \in Q$ on voyage $v \in V$
$\bar{H}_{irqv}$	Auxiliary variable for the number of TEUs loaded/unloaded at port $i \in P$ with handling rate $r \in R$ from/to ship $q \in Q$ on voyage $v \in V$
X_{iqv}^{skip}	The binary variable that equals 1 if port $i \in P$ for ship $q \in Q$ is skipped on voyage $v \in V$, otherwise equals 0
K_{iqv}^{skip}	The auxiliary variable for X_{iqv}^{skip}
G_{iqv}	The auxiliary variable for robust counterpart
W_{iqv}	The auxiliary variable for robust counterpart
θ_{iqv}	The auxiliary variable for robust counterpart
Z_{ijqv}	Auxiliary binary variable for demand at a skipped port
B_{iiqv}	Auxiliary binary variable for demand at a skipped port
E_{iiqv}	Auxiliary binary variable for demand at a skipped port
L_{ijqv}	Auxiliary binary variable for demand at a skipped port
A_{iiqv}	Auxiliary binary variable for demand at a skipped port
Div_{ikjqv}^{road}	Number of incoming TEUs to skipped port $j \in P$ from port $i \in P$ diverted from skipped port $j \in P$ to port $k \in P$ with ship $q \in Q$ on voyage $v \in V$ by road
Div_{ikjqv}^{sea}	Number of incoming TEUs to skipped port $j \in P$ from port $i \in P$ diverted from skipped port $j \in P$ to port $k \in P$ with ship $q \in Q$ on voyage $v \in V$ to be transported to the skipped port by sea
Div_{ijqv}^{round}	Number of incoming TEUs to skipped port $j \in P$ from port $i \in P$ diverted from skipped port $j \in P$ with ship $q \in Q$ on voyage $v \in V$ to be transported to the skipped port in the next voyage
I_{ijqv}^{out}	Number of outgoing containers held at skipped port $i \in P$ to be transported to port $j \in P$ with ship $q \in Q$ on voyage $v \in V$
I_{ijqv}^{wau}	Number of containers held at skipped port $i \in P$ to be transported to port $j \in P$ with ship $q \in Q$ on voyage $v \in V$
XR_{iqv}	Number of TEUs remain on ship in leg $i \in P$ on ship $q \in Q$ on voyage $v \in V$
S_{iqv}	Average speed in leg $i \in P$ of ship $q \in Q$ on voyage $v \in V$
U_{iqv}	Reciprocal average speed in leg $i \in P$ of ship $q \in Q$ on voyage $v \in V$
U_{iqvs}	Reciprocal average speed in leg $i \in P$ of ship $q \in Q$ for line $s \in L$ on voyage $v \in V$
Y_{iqvs}	The binary variable that equals 1 if line $s \in L$ in leg $i \in P$ for ship $q \in Q$ is selected on voyage $v \in V$, otherwise equals 0
F_{iqv}	Bunker consumption per km in leg $i \in P$ of ship $q \in Q$ on voyage $v \in V$
$\bar{F}_{ijqv}$	Auxiliary variable for bunker consumption per km in sea passage between port $i \in P$ and port $j \in P$ of ship $q \in Q$ on voyage $v \in V$ if the sea passage conducted
T_{iqv}^{sail}	Sail time in leg $i \in P$ of ship $q \in Q$ on voyage $v \in V$
$T_{iqv}^{sail_sec}$	Sail time in leg $i \in P$ of ship $q \in Q$ on voyage $v \in V$ if the next port(s) in the sequence of the service is not skipped
$T_{iiqv}^{sail_skip}$	Sail time from port $i \in P$ to $j \in P$ for ship $q \in Q$ on voyage $v \in V$ if the next port(s) in the sequence of the service is skipped
T_{iqv}^{arr}	Arrival time at port $i \in P$ of ship $q \in Q$ on voyage $v \in V$
T_{iqv}^{dep}	Departure time from port $i \in P$ of ship $q \in Q$ on voyage $v \in V$
T_{irqv}^{hand}	Container handling time at port $i \in P$ of ship $q \in Q$ on voyage $v \in V$
T_{iqv}^{wait}	Waiting time at port $i \in P$ of ship $q \in Q$ on voyage $v \in V$
T_{iqv}^{late}	Late arrival time at port $i \in P$ of ship $q \in Q$ on voyage $v \in V$
T_{qv}^{slup}	Auxiliary variable for the total time of ship $q \in Q$ on voyage $v \in V$
X_{iqv}^{late}	The binary variable that equals 1 if ship $q \in Q$ on voyage $v \in V$ arrives late to port $i \in P$, otherwise equals 0

found that collaborative agreements with terminal operators can reduce total costs as well as emissions.

B. DISRUPTION RECOVERY

Disruptions in container shipping vessel schedules are inevitable. They occur either from delays in sea passages or from delays in ports of a container line service that vessels operate in. They usually create financial loss because of customer dissatisfaction and/or cost increases resulting from recovery actions. Schedule recovery from disruptions with speeding up not only increases bunker expenses but also increases the carbon footprints of container lines. Brouer et al. [27] revealed that application of recovery measures i.e., speed adjustment, port skipping, and port swapping may lead to cost savings of up to 58%. Findings of Cheraghchi et al. [28] showed that the disruptions in long-distance sea voyages are easy to mitigate by speed adjustments compared with short distant sea voyages. On the other hand, Li et al. [29] found that speeding up may not be effective recovery option for long disruptions compared with port skipping. Abioye et al. [30] suggested that long disruptions require port skipping and other measures such as higher handling rates. The results of the study by Abioye et al. [31] showed that usage of disruption recovery options depends on their costs, speed adjustment can be costly when fuel costs are high and port skipping is also highly dependent on its associated costs. In addition, De et al. found that [32] disruptions recovery measures can be integrated with bunker management through vessel rescheduling to create additional cost savings.

C. UNCERTAINTY IN PORT AND SAILING TIMES

Most of the time, occurrences and durations of disruptions cannot be predicted. Therefore, to provide robust vessel schedules, the modeling approach must account for uncertainties in port times and sailing times. By accounting for uncertainties, cost savings can be achieved. Additionally, fuel costs can be reduced by sacrificing the service levels. However, Song et al. [26] showed that those service levels can also be improved considerably by enduring small costs. Wang and Meng [38] found that one way to hedge against the uncertainties in port operations is adding more ships to container line services. On the other hand, findings of Wang and Meng [37] and Aydin et al. [40] underlined that uncertainties in port times, as well as sailing times, can also be dealt with increasing the optimal speed in each leg of a service rather than increasing the number of vessels assigned to it. In addition, Ma et al. [43] revealed that fuel consumption resulting from the uncertainty of port times can be reduced by increasing the availability of port time windows. Moreover, findings of Li et al. [39] suggested that adopting real-time disruption recovery policies helps reduce the costs incurred from uncertain port times. Mulder and Dekker [42] showed that another way to deal with port and sailing time uncertainties is buffer time allocation which can decrease costs around 28.9%.

D. COLLABORATION WITH TERMINAL OPERATORS

Disruptions in vessel schedules are mostly recovered by operational measures such as speeding up and port skipping. Speeding up not only increases bunker consumption costs but also increases the carbon footprints of container lines. Port skipping also causes problems because container transportation demand from/to the skipped ports must be diverted to other ports in the container line service with additional cost and transit time increases. As another measure for recovering from disruptions, container lines and container terminal operators can collaborate especially regarding the provision of arrival time windows and container handling rates. Collaboration for port time windows to increase the availability of port time windows can reduce total costs [7] and the optimal number of ships deployed [19]. Liu et al. [35] proposed a collaborative mechanism where port operators provide shorter port times and container lines compensate the port operators can help reduce costs for container lines without incurring additional costs to port operators. Dulebenets [15] and Dulebenets [33] found that collaborating with terminal operators not only for the availability of port time windows or handling rates provision but for both of them can even further reduce costs compared to collaborating for only one of them. Additionally, findings of Venturini et al. [34] underlined that collaborating for berth allocation between container lines and port operators can lead to more efficient and equitable berth allocation plans compared with the alternative without collaboration. Dulebenets [8] showed that collaborative agreements, i.e., handling rate and port time window provision, not only can reduce costs but also can help to reduce the emissions produced by a container line service.

E. CONTRIBUTION OF THE STUDY

Uncertainties in ports and sailings can create disruptions that require recovery measures such as collaboration with terminal operators, port skipping and speeding up. Even though those factors significantly impact each other, Table 1 illustrates that none of the previous studies considered them all together. In this regard, this study investigates the question that how robust container line service schedules which minimize cost and emissions can be designed taking advantage of collaboration with terminal operators as well as operational measures i.e., port skipping and speed adjustment. This study provides a comprehensive modeling approach that was not provided by the previous studies by proposing a multi-objective mixed-integer non-linear robust programming model. The model handles tactical schedule design for any container line service. Rare and big disruptions best handled by operational and/or real time models and those kinds of models proposed in the previous studies [27], [29], [39]. The contribution of this study can be summarized as follows:

1. The model includes disruptions recovery, collaboration with terminal operators, and uncertainties in port and sailing times in vessel schedules. Three aspects were

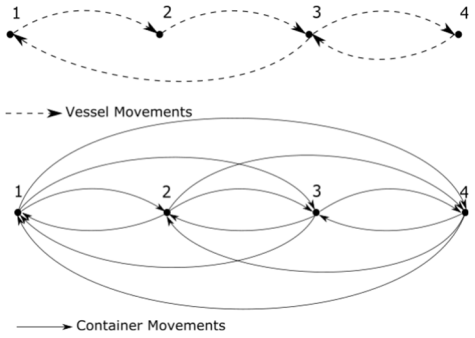


FIGURE 1. A container line service.

included together in the model because they impact each other. Uncertainties in the port and sailing times caused by factors such as crane shutdown, worker strikes, or engine broke down at sea can create disruptions. Unless some recovery measures are taken, those disruptions can affect the whole schedule of a container line service. Speeding up and port skipping are two of the widely used operational measures for disruption recovery. However, speeding up increases fuel costs and the carbon footprint of container lines and port skipping also incur an additional cost of container divergence. Collaborating with terminal operators regarding available time windows and handling rates can help to design green and economic container liner service schedules.

2. The model proposed in this study includes multi-voyages as multi-periods and separately modeled multiple ships that were not considered in the previous studies. Explicit modeling for multi-ships can affect recovery decisions, for example, one ship can only unload containers in a port and the containers to be loaded on the ship at that port can wait for the next ship. Multi-voyages as a multi-period approach can also impact the robustness of decisions because container shipping schedules are designed for around six months to one year in practice.
3. The objective functions of the model include not only costs but also emissions with the adoption of a multi-objective approach. GHG emission caused by ships is currently under the scrutiny of authorities. Mathematical modeling can provide an effective tool for reducing those emissions [24], [42].
4. Apart from all that, an algorithm was provided to linearize the non-linear bunker consumption function that allows controlling the error term.

III. PROBLEM DESCRIPTION

This study deals with the problem of green container shipping vessel scheduling. Vessel scheduling decisions concern each route of a container line. Container lines operate in fixed routes, which are called container line services. Figure 1 illustrates a container line service with four ports $P = \{1, 2, 3, 4\}$. The vessels operating in a container service are dedicated to only one service. Vessels operate sequentially in a cyclic way visiting each port of the service. However, if there is a disruption at a port, vessels can skip that port. Containers coming to or going out of the skipped port can be diverted to another port which has road connection to the skipped port or diverted by sea with transshipment. The unloaded containers at the diverted port can be transported to the skipped port by road or by sea with another service of the container line or by a service of another container line. The same situation applies to outgoing containers from the skipped port.

If it is assumed that port 1 is the first port of the service illustrated in Figure 1, a vessel performs a voyage that starts from port 1 and visits port 2, port 3, and port 4 and then comes back to port 1 as a round trip. This is repeated for each voyage. Additionally, the sea passage between two ports in a container line service is called a leg.

Figure 1 illustrates the movements of vessels and containers in a container service. It can be assumed that port 1 is the first port of the service. When a vessel arrives at port 2, it unloads containers that are coming from port 1 in the current voyage and unloads containers loaded to the vessel in the previous voyage at port 3 and port 4. And the vessel loads containers at port 2 to be delivered at port 3 and port 4 in the current voyage and to be delivered at port 1 in the next voyage. When a vessel arrives at a port, the containers loaded at the previous ports to be delivered at the upcoming ports always remain on the vessel. This process is replicated for each port and each voyage. The vessel operations in those ports and the sea passages between them do not always go smoothly; disruption and delays are inevitable. When a disruption occurs, it has the potential to impact the schedules of every ship operating in a container line service. Therefore, vessels in a container line service need to recover from disruptions. For vessel schedule recovery, container lines can implement various operational measures e.g., speed adjustments and port skipping. When a port is skipped, the containers to be transported to/from it must be diverted to another port. In addition to operational measures, a container line can plan robust green vessel schedules by collaborating with terminal operators in terms of container handling rates and/or time window availability by sharing real-time information regarding estimated/actual port times and sea passage times and estimated/actual disruption durations at ports and sea passages.

IV. MODEL DESCRIPTION

In this section, a comprehensive robust multi-objective mixed-integer non-linear programming model was proposed. Table 2 shows the model nomenclature that includes sets, parameters, and decision variables.

Expressions 1.1-1.9 incorporates the first objective function that minimizes total operational costs. Expression 1.1 is the total fuel consumption cost. Expression 1.2 is the total container handling cost. Expressions 1.3-1.7 are the costs of

container diversion in case of port skipping. Expression 1.7 is the total late arrival cost for ships, and Expression 1.9 is the total ship operating costs.

$$\min z_1 = c^{fuel} \sum_{i\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} \hat{F}_{ijqv}\, dist_{ij} + \tag{1.1}$$

$$\sum_{i\in P}\sum_{r\in R}\sum_{q\in Q}\sum_{v\in V} H^{aux}_{irqv}\, c^{hand}_{r} + \tag{1.2}$$

$$\sum_{i\in P}\sum_{k\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} Div^{road}_{ikjqv}\, c^{road_div}_{kj} + \tag{1.3}$$

$$c^{sea_div} \sum_{i\in P}\sum_{k\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} Div^{sea}_{ikjqv} + \tag{1.4}$$

$$c^{round} \sum_{i\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} Div^{round}_{ijqv} + \tag{1.5}$$

$$c^{stor}_{i} \sum_{i\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} I^{out}_{ijqv} + \tag{1.6}$$

$$c^{stor}_{i} \sum_{i\in P}\sum_{j\in P}\sum_{q\in Q} I^{hold}_{ijqv} + \tag{1.7}$$

$$\sum_{i\in P}\sum_{q\in Q}\sum_{v\in V} T^{late}_{iqv}\, c^{late}_{i}/24 + \tag{1.8}$$

$$\sum_{q\in Q}\sum_{v\in V} T^{ship}_{qv}\, c^{opr}_{q}/24 \tag{1.9}$$

Expressions 2.1-2.3 are the second objective function that minimizes total emissions costs. Expression 2.1 is the total emissions resulting from fuel consumption. Expression 2.2 is the total emissions produced by container handling. Expression 2.3 is the emissions produced from container diversion in cases of port skipping. Total emissions cost equals total emission tonnage multiplied by the carbon price.

$$\min z_2 = c^{ems} \Bigg(e^{fuel} \sum_{i\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} \hat{F}_{ijqv}\, dist_{ij} + \tag{2.1}$$

$$e^{hand} \sum_{i\in P}\sum_{q\in Q}\sum_{v\in V} H_{iqv} \tag{2.2}$$

$$\sum_{i\in P}\sum_{k\in P}\sum_{j\in P}\sum_{q\in Q}\sum_{v\in V} Div^{road}_{ikjqv}\, e^{road_div}_{kj} \Bigg) \tag{2.3}$$

Equations 3.1-3.3 determine the number of containers to be transported between each port. Two equations are identical except Equations 3.2 and 3.3 were included to keep the continuum. The expression on the left-hand side represents the number of containers to be transported from port i to port j by ship q on voyage v. The first expression on the right-hand side of the equation is the container transportation demand from port i to port j for ship q on voyage v. The demand parameter in the expression was multiplied by a binary variable to force that when port i and/or port j is skipped, container transportation from and to the skipped port(s) would not occur. The expression is accompanied by Equations 4-7. The second, third, and fourth expressions on the right-hand side of the Equations are the numbers of incoming containers diverted from skipped ports to another port to be transported to the skipped ports by road; the number of incoming containers diverted to another port to be transported to the skipped port by the sea with transshipping to another ship of either another service of the container line or service of another container line; the number of incoming diverted containers to be transported to destination ports by the same ship in the next voyage, respectively. And the fifth expression is the number of outgoing containers that were waiting at skipped ports.

$$X_{ijqv} = d_{ijqv}\left(1 - E_{ijqv}\right) + \sum_{k\in P} Div^{road}_{ijkqv} + \sum_{k\in P} Div^{sea}_{ijqv} + Div^{round}_{ijqv} + I^{out}_{ijqv}, \quad \forall i \in P,\ \forall j \in P, i \le j, \forall q \in Q,\ \forall v \in V \tag{3.1}$$

$$X_{ijq,v+1} = d_{ijqv}\left(1 - E_{ijqv}\right) + \sum_{k\in P} Div^{road}_{ijkqv} + \sum_{k\in P} Div^{sea}_{ijkqv} + Div^{round}_{ijqv} + I^{out}_{ijqv}, \quad \forall i \in P,\ \forall j \in P, i > j, \forall q \in Q,\ \forall v \in V\backslash\{|V|\} \tag{3.2}$$

$$X_{ijq,\min V} = d_{ijqv}\left(1 - E_{ijqv}\right) + \sum_{k\in P} Div^{road}_{ijkqv} + \sum_{k\in P} Div^{sea}_{ijkqv} + Div^{round}_{ijqv} + I^{out}_{ijqv}, \quad \forall i \in P,\ \forall j \in P, i > j,\ \forall q \in Q,\ v = |V| \tag{3.3}$$

Equations 4-10 regulate the logic of container flows from/to skipped ports. There are four possibilities; the first one is that the outgoing port is skipped, and the incoming port is not, the second one is the opposite of the first one, the third one is both ports are skipped, and the fourth one is both ports are not skipped. The expression $\left(1 - E_{ijqv}\right)$ covers the possibility of both ports not being skipped by only being equal to 1 only if both outgoing and incoming ports are not skipped. The variable A_{ijqv} covers the possibility of the outgoing port is not skipped but the incoming port is skipped by being equal to 1 only if the outgoing port is not skipped and the incoming port is skipped. The variable B_{ijqv} covers two possibilities both ports are skipped, and the incoming port is skipped by being equal to 1 only if both ports are skipped or incoming ports are skipped. Equation 11 ensures that only the ports where disruption occurs can be skipped.

$$Z_{ijqv} = X^{skip}_{iqv}, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{4}$$

$$B_{ijqv} = X^{skip}_{jqv}, \quad \forall i \in P,\ \forall j \in P, i \le j, \forall q \in Q,\ \forall v \in V \tag{5.1}$$

$$B_{ijqv} = X^{skip}_{iq,v+1}, \quad \forall i \in P,\ \forall j \in P, i > j,\ \forall q \in Q, \forall v \in V\backslash\{|V|\} \tag{5.2}$$

$$B_{ijqv} = X^{skip}_{iq,\min V}, \quad \forall i \in P,\ \forall j \in P, i > j, \forall q \in Q, v = |V| \tag{5.3}$$

$$Z_{ijqv} + B_{ijqv} \ge E_{ijqv}, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{6}$$

$$Z_{ijqv} + B_{ijqv} \leq 2E_{ijqv}, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{7}$$

$$L_{ijqv} = 1 - Z_{ijqv}, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{8}$$

$$L_{ijqv} + B_{ijqv} \leq A_{ijqv} + 1, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{9}$$

$$L_{ijqv} + B_{ijqv} \geq 2A_{ijqv}, \quad \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{10}$$

$$X^{skip}_{iqv} \leq \delta^{port_occ}_{iqv}, \quad \forall i \in P,\ \forall q \in Q,\ \forall v \in V \tag{11}$$

Equations 12.1-12.4 ensure that when a port is skipped, incoming containers to that port must be diverted. The first and second expression on the left-hand side of the equation is the number of incoming containers to a skipped port diverted to a port that has a road connection to the skipped port. It can be seen that the diversion can be done to the ports that come after the skipped port and before the outgoing port in the sequence of the container line service. The diverted port can be the same as the outgoing port. It means that containers diverted to the skipped port can be directly transported by road if a road connection is available. The third and fourth expression on the left-hand side provides that incoming containers to a skipped port can be diverted by sea by another service of the container line or by a service of another container line. It can be noticed that the diverted port cannot be the same as the outgoing port. The last term on the left-hand side enables the incoming containers to a skipped port can be diverted to the skipped port by the same ship on the next voyage. The expression on the right-hand side of the equation forces that when the outgoing port is not skipped and the incoming port is skipped, incoming container transportation demand to a skipped port must be diverted. Equations 13-15.2 ensure that incoming containers cannot be diverted to a skipped port.

$$\sum_{\substack{k \in P \\ k > j}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ k \leq i}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ k > j}} Div^{sea}_{ikjqv} + \sum_{\substack{k \in P \\ k < i}} Div^{sea}_{ikjqv} + Div^{round}_{ijq,v+1} = d_{ijqv}A_{ijqv},\ \forall i \in P,\ \forall j \in P, i \leq j,\ \forall q \in Q,\ \forall v \in V\backslash\{|V|\} \tag{12.1}$$

$$\sum_{\substack{k \in P \\ k > j}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ k \leq i}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ k > j}} Div^{sea}_{ikjqv} + \sum_{\substack{k \in P \\ k < i}} Div^{sea}_{ikjqv} + Div^{round}_{ijq,\min V} = d_{ijqv}A_{ijqv},\ \forall i \in P,\ \forall j \in P, i \leq j,\ \forall q \in Q,\ v = |V| \tag{12.2}$$

$$\sum_{\substack{k \in P \\ j < k \leq i}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ j < k < i}} Div^{sea}_{ikjqv} + Div^{round}_{ijq,v+1} = d_{ijqv}A_{ijqv},\ \forall i \in P,\ \forall j \in P, i > j,\ \forall q \in Q,\ \forall v \in V\backslash\{|V|\} \tag{12.3}$$

$$\sum_{\substack{k \in P \\ j < k \leq i}} Div^{road}_{ikjqv}\delta^{road_div}_{kj} + \sum_{\substack{k \in P \\ j < k < i}} Div^{sea}_{ikjqv} + Div^{round}_{ijq,\min V} = d_{ijqv}A_{ijqv},\ \forall i \in P,\ \forall j \in P, i > j,\ \forall q \in Q,\ v = |V| \tag{12.4}$$

$$Div^{road}_{ikjqv} \leq d_{ijqv}\left(1 - E_{ikqv}\right),\ \forall i \in P,\ \forall k \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{13}$$

$$Div^{sea}_{ikjqv} \leq d_{ijqv}\left(1 - E_{ikqv}\right),\ \forall i \in P,\ \forall k \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{14}$$

$$Div^{round}_{ijqv} \leq d_{ijq,v-1}\left(1 - E_{ijqv}\right),\ \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V\backslash\{\min V\} \tag{15.1}$$

$$Div^{round}_{ijqv} \leq d_{ijq,|Q|}\left(1 - E_{ijqv}\right),\ \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ v = \min V \tag{15.2}$$

Equation 16.1-16.3 forces that when an outgoing port or both outgoing port and incoming port are skipped, outgoing containers from the outgoing skipped port must be held at the outgoing port. The first expression on the left-hand side of the equation is the number of outgoing containers from a skipped port held at the outgoing port for the next ship(s). The second expression on the left-hand side is the outgoing containers that were held one more week at the outgoing port for the next ship(s). The first expression on the right-hand side of the equation forces that when an outgoing port or both outgoing port and incoming port are skipped, those containers must be diverted. The second expression is the number of containers held at the outgoing port for the next ship(s). Equation 17 ensures that containers cannot be diverted to a skipped port.

$$I^{out}_{ijqv} + I^{hold}_{ijqv} = d_{ij,q-1,v}Z_{ij,q-1,v} + I^{hold}_{ij,q-1,v},\ \forall i \in P,\ \forall j \in P,\ \forall q \in Q\backslash\{\min Q\},\ \forall v \in V \tag{16.1}$$

$$I^{out}_{ijqv} + I^{hold}_{ijqv} = d_{ij,|Q|,v}Z_{ij,|Q|,v} + I^{hold}_{ij,\ |Q|,v},\ \forall i \in P,\ \forall j \in P,\ q = minQ,\ \forall v \in V\backslash\{\min V\} \tag{16.2}$$

$$I^{out}_{ijqv} + I^{hold}_{ijqv} = d_{ij,\ |Q|,\ |V|}Z_{ij,\ |Q|,\ |V|} + I^{hold}_{ij,|Q|,\ |V|},\ \forall i \in P,\ \forall j \in P,\ q = minQ,\ v = minV \tag{16.3}$$

$$I^{out}_{ijqv} \leq d_{ijqv}\left(1 - E_{ijqv}\right),\ \forall i \in P,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{17}$$

Equations 18.1-19.2 regulate ship capacity. The first term on the left-hand side of Equations 18.1 and 18.2 is the number of containers that remain on a ship after the ship was unloaded at a port. The first term on the right-hand side is the sum of the number of containers to be transported from the origin ports that come before the arrived port in the sequence of the container line service to the destination ports that come after the arrived port in the sequence of the container line

service. The second term on the right-hand side is the sum of the number of containers to be transported from the origin ports that come before the arrived port in the sequence of the service to the destination ports that come before the origin ports in the sequence of the container line service. The last term on the right-hand side of the equation is the sum of the number of containers to be transported from the origin ports that come after the arrived port in the sequence of the service to the destination ports that come before the origin ports in the sequence of the container line service. Equations 19.1-19.2 ensures that the number of containers loaded on a ship at a port cannot be more than the capacity of the ship minus the number of containers that remain on the ship. The term on the left-hand side of Equations is the sum of the number of containers loaded on a ship. The first term on the right-hand side of the equation is the capacity of the ship and the second term on the right-hand side of the equation is the number of containers that remain on the ship.

$$XR_{iqv} = \sum_{\substack{k\in P\\ k>i}}\sum_{\substack{j\in P\\ i>j}} X_{jkqv} + \sum_{\substack{j\in P\\ i>j}}\sum_{\substack{k\in P\\ j>k}} X_{jkq,v+1} + \sum_{\substack{j\in P\\ j>k}}\sum_{\substack{k\in P\\ k>i}} X_{jkq,v+1},\ \forall i\in P,\ \forall q\in Q,\ \forall v\in V\setminus\{|V|\} \tag{18.1}$$

$$XR_{iqv} = \sum_{\substack{k\in P\\ k>i}}\sum_{\substack{j\in P\\ i>j}} X_{jkqv} + \sum_{\substack{j\in P\\ i>j}}\sum_{\substack{k\in P\\ j>k}} X_{jkq,\min V} + \sum_{\substack{j\in P\\ j>k}}\sum_{\substack{k\in P\\ k>i}} X_{jkq,\min V},\ \forall i\in P,\ \forall q\in Q,\ v=|V| \tag{18.2}$$

$$\sum_{\substack{j\in P\\ j>i}} X_{ijqv} + \sum_{\substack{j\in P\\ j<i}} X_{ijq,v+1} \le cap_q - XR_{iqv},\ \forall i\in P,\ \forall q\in Q,\ \forall v\in V\setminus\{|V|\} \tag{19.1}$$

$$\sum_{\substack{j\in P\\ j>i}} X_{ijqv} + \sum_{\substack{j\in P\\ j<i}} X_{ijq,\min V} \le cap_q - XR_{iqv},\ \forall i\in P,\ \forall q\in Q,\ v=|V| \tag{19.2}$$

Equation 20 and 21 regulate bunker consumption. Bunker consumption per day of a container ship is assumed to have a power relation with the speed of the ship [44]. γ and α are the coefficients that define that relationship. Thus, daily bunker consumption equals $\gamma * s^a$, where s is the daily average speed. On the other hand, Equation 20 regulates the bunker consumption of a ship per mile. Therefore, the equation was converted to per-mile bunker consumption by dividing it by daily mileage ($\gamma * s^a/(s*24) = \gamma * s^{a-1}/24$). The term on the left-hand side of Equation 20 is the bunker consumption per mile of a ship on leg $i \in P$. The term on the right-hand side of Equation 20 is the average speed of a ship in leg $i \in P$ with the bunker consumption coefficients which define its relation. Equation 21 regulates bunker consumption in sea passage from port $i \in P$ to port $j \in P$ if the sea passage is conducted. The term on the left-hand side is the auxiliary variable for per-mile bunker consumption it equals bunker consumption per mile multiplied by the binary variable that equals 1 if the sea passage is conducted between port $i \in P$ and $j \in P$. It can be seen that both Equations 20 and 21 contain non-linear terms and they were linearized in the upcoming sections.

$$F_{iqv} = \gamma S_{iqv}^{\alpha-1} \div 24,\ \forall i\in P,\ \forall q\in Q,\ \forall v\in V \tag{20}$$

$$\hat{F}_{ijqv} = F_{iqv} XY_{ijqv},\ \forall i\in P,\ \forall j\in P,\ \forall q\in Q,\ \forall v\in V \tag{21}$$

Equations 22-24.2 determine the sailing times of ships. The term on the left-hand side of Equation 22 is the sailing time of a ship in leg $i \in P$ and the first term on the right-hand side is the sailing time in leg $i \in P$ if the sequential sea passages were conducted. The second term on the right-hand side is the sailing time if the next port(s) of port $i \in P$ was skipped. Equations 23.1 and 23.2 regulate sailing times in sequential sea passages. The term on the left-hand side is the sailing time of a ship in sea leg $i \in P$. The first term on the right-hand side is the distance between port $i \in P$ and $i+1 \in P$ divided by the average speed in the sea leg. The second and third terms on the right-hand side are the disruption durations if occurs on sea leg $i \in P$ and strait passage duration if the sea leg has a strait passage. The last term on the right-hand side ensures that the sailing time is only applicable if the sea passage between port $i \in P$ and $i+1 \in P$ is conducted. Equations 24.1 and 24.2 regulate sailing times if the next port(s) of port $i \in P$ was skipped. The term on the left-hand side is the sailing time between port $i \in P$ and $j \in P$ where sea passage between port $i \in P$ and $j \in P$ is conducted. The first term on the right-hand side is the distance between port $i \in P$ and $j \in P$ divided by the average speed in the sea leg. The second and third terms on the right-hand side are the disruption durations if occurred on sea leg $i \in P$ and strait passage duration if the sea leg has a strait passage. The fourth term on the right-hand side ensures that the sailing time is only applicable if the sea passage between port $i \in P$ and $j \in P$ is conducted. The last term ensures that if sea passage between port $i \in P$ and $i+1 \in P$ is conducted the sailing time from port $i \in P$ to port $j \in P$ is not applicable.

$$T_{iqv}^{sail} = T_{iqv}^{sail_sec} + \sum_{j\in P} T_{ijqv}^{sail_skip},\ \forall i\in P,\ \forall q\in Q,\ \forall v\in V \tag{22}$$

$$T_{iqv}^{sail_sec} \ge dist_{i,i+1} \div S_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,i+1}^{str_pass}\delta_{i,i+1}^{str_dur} - (1-XY_{i,i+1,qv})M,\ \forall i\in P\setminus\{|P|\},\ \forall q\in Q,\ \forall v\in V \tag{23.1}$$

$$T_{iqv}^{sail_sec} \ge dist_{i,minP} \div S_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,\min P}^{str_pass}\delta_{i,\min P}^{str_dur} - (1-XY_{i,minP,q}v)M,\ i=|P|,\ \forall q\in Q,\ \forall v\in V \tag{23.2}$$

$$T^{sail_skip}_{ijqv} \geq dist_{ij} \div S_{iqv} + \delta^{sea_occ}_{iqv}\xi^{sea_dur}_{iqv} + \delta^{str_pass}_{ij}\delta^{str_dur}_{ij} - (1 - XY_{ijqv})M - XY_{i,i+1,qv}M, \ \forall i \in P\backslash\{|P|\}, \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (24.1)$$

$$T^{sail_skip}_{ijqv} \geq dist_{ij} \div S_{iqv} + \delta^{sea_occ}_{iqv}\xi^{sea_dur}_{iqv} + \delta^{str_pass}_{ij}\delta^{str_dur}_{ij} - (1 - XY_{ijqv})M - XY_{i,minP,qv}M, \ i = |P|, \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (24.2)$$

Equations 25.1- 27.2 regulate sequential sea passages between ports. Equations 25.1 and 25.2 ensure that if one of the ports or both in the sequential sea passages, which are the passages between port $i \in P$ and $i+1 \in P$, is skipped, the sea passage is not conducted. Equation 26 ensures that a sea passage from a port is only conducted once in a voyage of a ship and if the port is skipped, the sea passage from the skipped port is not conducted at all. Equations 27.1 and 27.2 are the flow conservation constraints that ensure the number of sea passages going out of a port equals the number of passages going into a port.

$$\left(1 - E_{i,i+1,qv}\right) = XY_{i,i+1,qv}, \ \forall i \in P\backslash\{|P|\}, \forall q \in Q, \forall v \in V \quad (25.1)$$

$$\left(1 - E_{i,minP,qv}\right) = XY_{i,minP,qv}, i = |P|, \ \forall q \in Q, \ \forall v \in V \quad (25.2)$$

$$\sum_{\substack{j \in P \\ j \neq i}} XY_{ijqv} = 1 - X^{skip}_{iqv}, \ \forall i \in P, \ \forall q \in Q, \ \forall v \in V \quad (26)$$

$$\sum_{\substack{j \in P \\ j \neq i}} XY_{ijqv} = \sum_{\substack{j \in P \\ j < i}} XY_{jiqv} + \sum_{\substack{j \in P \\ j > i}} XY_{jiq,v-1}, \ \forall i \in P, \forall q \in Q, \ \forall v \in V\backslash\{\min V\} \quad (27.1)$$

$$\sum_{\substack{j \in P \\ j \neq i}} XY_{ijqv} = \sum_{\substack{j \in P \\ j < i}} XY_{jiqv} + \sum_{\substack{j \in P \\ j > i}} XY_{jiq,|V|}, \ \forall i \in P, \forall q \in Q, \ v = \min V \quad (27.2)$$

Equations 28.1- 28.3 determine the arrival times of ships. It can be noticed that three equations are the derivation of one. Because there is no port before the first port in a sequence of a container line service, Equations 28.2 and 28.3 ensures that the last port comes before the first port. The term on the left-hand side of the equation is the arrival time of a ship to a port. The first term on the right-hand side of the equation is the departure time of a ship from the preceding port. And the second term on the right-hand side is the sailing duration on the leg before the arrival port. The last term on the right-hand side is the total duration of the voyage for the ship. It is subtracted from the sum of the departure time and the sailing duration of the last leg to regulate the arrival time of the first leg of a ship on a voyage.

$$T^{arr}_{iqv} = T^{dep}_{i-1,qv} + T^{sail}_{i-1,qv}, \ \forall i \in P\backslash\{\min P\}, q \in Q, \ \forall v \in V \quad (28.1)$$

$$T^{arr}_{iqv} = T^{dep}_{|P|,q,v-1} + T^{sail}_{|P|,q,v-1} - T^{ship}_{q,v-1}, \ i = \min P, q \in Q, \ \forall v \in V\backslash\{\min V\} \quad (28.2)$$

$$T^{arr}_{iqv} = T^{dep}_{|P|,q,|V|} + T^{sail}_{|P|,q,|V|} - T^{ship}_{q,|V|}, i = \min P, \ q \in Q, \ v = \min V \quad (28.3)$$

Equation 29 regulates the waiting time of ships at each port if a ship arrives at a port before its time window has not started. The term on the left-hand side of the equation is the wait duration. The first term on the right-hand side of the equation is the start time of the time window. The second term is the arrival time of the ship, and it is subtracted from the start time of the time window. If the result of the subtraction is positive, it means that the ship arrived at the port early and must wait. The last term ensures that when the port is skipped, no waiting time is incurred.

$$T^{wait}_{iqv} \geq Tw^{st}_{iqv} - T^{arr}_{iqv} - MX^{skip}_{iqv}, \ \forall i \in P, \ \forall q \in Q, \ \forall v \in V \quad (29)$$

Equation 30 regulates the late arrival of ships. The term on the left-hand side of Equation 30 is the late arrival duration. The first term on the right-hand side of the equation is the arrival time of a ship and the second term is the end time of the arrival time window. The end time of the arrival window is subtracted from the arrival time to determine the late arrival duration. The third term on the right-hand side of the equation ensures the late arrival duration does not incur for the ports that were skipped. Equation 31 ensures that the time window duration equals the time window start time subtracted from the time window end time. Equation 32 determines if a ship arrived late at a port or not. This equation is included because when a ship arrives late at a port, loading and discharging operations do not start right away unless there is available berth space. Therefore, most of the time in the case of late arrival, the ship must wait for an available berth space and that causes an additional delay.

$$T^{late}_{iqv} \geq T^{arr}_{iqv} - Tw^{end}_{iqv} - MX^{skip}_{iqv}, \ \forall i \in P, \forall q \in Q, \ \forall v \in V \quad (30)$$

$$Tw^{end}_{iqv} - Tw^{st}_{iqv} \leq tw^{dur}, \ \forall i \in P, \ \forall q \in Q, \ \forall v \in V \quad (31)$$

$$T^{late}_{iqv} \leq M^{late}X^{late}_{iqv}, \ \forall i \in P, \ \forall q \in Q, \ \forall v \in V \quad (32)$$

$$X^{late}_{iqv} \leq 1 - X^{skip}_{iqv}, \ \forall i \in P, \ \forall q \in Q, \ \forall v \in V \quad (33)$$

Equation 34 determines the departure times of ships. The term on the left-hand side is the departure time and it equals the sum of arrival time, waiting time, container handling time, delay durations, and delays incurred from late arrivals. It can be seen that Equation 35 is created to make sure the delay duration equals zero when the port is skipped. Equation 36 regulates the port disruption duration. The terms on the left-hand side of Equation 36 and Equation 35 could be directly included in Equation 34, however, it is constructed separately for robust counterpart reformulation (further details are in the

related section).

$$T^{dep}_{iqv} \geq T^{arr}_{iqv} + T^{wait}_{iqv} + \sum_{r\in R} T^{hand}_{irqv} + \delta^{port_occ}_{iqv} G_{iqv} + t^{fix}_{i} K^{skip}_{iqv} + \delta^{late}_{i} X^{late}_{iqv},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \quad (34)$$

$$K^{skip}_{iqv} = 1 - X^{skip}_{iqv},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \quad (35)$$

$$G_{iqv} \geq \xi^{port_{dur}}_{iqv} K^{skip}_{iqv},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \quad (36)$$

Equations 37.1 and 37.2 determine the number of containers to be handled at a port. The first term on the left-hand side of the equation is the total number of outgoing containers loaded on board a ship at a port. It can be noticed that shipments that have loading and discharging ports the same are not included. Container transportation between ports where the origin and destination are the same only occurs when there is a diversion, and it must not be counted in the number of containers loaded and discharged to/from a ship. The second term is the total number of incoming containers unloaded from a ship at a port. Equation 38 determines which handing rate is used to handle containers at a port. The number of containers handled at each port is included in the model as a decision variable with container handling rates and each handling rate has a different unit container handling cost. The equation forces that if the handling rate was not chosen, the number of containers handled at that rate equals zero otherwise equals the binary variable of chosen handling rate times the number of containers handled at that port at a voyage of a ship in the service. Equation 39 forces that only one container handling rate is selected at each port. Equation 38 contains a bilinear term, and it is linearized in the related section.

$$\sum_{\substack{j\in P\\ j\neq i}} X_{ijqv} + \sum_{\substack{j\in P\\ j<i}} X_{jiqv} + \sum_{\substack{j\in P\\ j>i}} X_{jiq,v+1} = H_{iqv},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V\backslash\{|V|\} \quad (37.1)$$

$$\sum_{\substack{j\in P\\ j\neq i}} X_{ijqv} + \sum_{\substack{j\in P\\ j<i}} X_{jiqv} + \sum_{\substack{j\in P\\ j>i}} X_{jiq,\min V} = H_{iqv},\ \forall i \in P,\ \forall q \in Q,\ v = |V| \quad (37.2)$$

$$\hat{H}_{irqv} = H_{iqv} * H^{rate}_{irqv},\ \forall i \in P,\ \forall r \in R,\ \forall q \in Q,\ \forall v \in V \quad (38)$$

$$\sum_{r\in R} H^{rate}_{irqv} = 1,\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \quad (39)$$

Equation 40 determines container handling times at ports. The term on the left-hand side is the container handling duration at a port. It forces that if the handling rate was not chosen, container handling duration equals zero otherwise equals the number of containers handled at that port divided by the unit handling duration of the chosen handling rate. As ensured in Equation 39, only one container handling rate is selected at each port. Equation 41 regulates the total duration of ships for all voyages. The term on the left-hand side is the total duration of a ship. The terms on the right-hand side of the equation are total sailing duration, total waiting duration, total container handling duration, port delay durations if occurred, and delays incurred from late arrivals of ships. Equation 42 ensures that the total duration of a ship's voyage must be less than or equal to the predetermined ship frequency multiplied by the predetermined number of ships in the service and it is multiplied by 24 to convert days into hours.

$$T^{hand}_{irqv} = H^{aux}_{irqv} \div h^{rate}_{r},\ \forall i \in P,\ \forall r \in R,\ \forall q \in Q,\ \forall v \in V \quad (40)$$

$$T^{ship}_{qv} \geq \sum_{i\in P} T^{sail}_{iqv} + \sum_{i\in P} T^{wait}_{iqv} + \sum_{i\in P}\sum_{r\in R} T^{hand}_{irqv} + \sum_{i\in P} \delta^{port_occ}_{iqv} G_{iqv} + \sum_{i\in P} t^{fix}_{i} K^{skip}_{iqv} + \sum_{i\in P} \delta^{late}_{i} X^{late}_{iqv},\ q \in Q,\ \forall v \in V \quad (41)$$

$$T^{ship}_{qv} \leq 24 * f * |Q|,\ q \in Q,\ \forall v \in V \quad (42)$$

A. LINEAR APPROXIMATION OF BUNKER CONSUMPTION FUNCTION

Equation 20 demonstrates the non-linear bunker consumption function that has a power relation with the speed of the ship [44]. To ease the solving process of the application instances, the power function of the bunker consumption function is required to be linearly approximated. Equations 43-45.2 demonstrate a new variable U_{iqv} that equals the reciprocal of speed ($1/S_{iqv}$). It was introduced in the model as a part of the linearization process. To incorporate the new variable in the model, Equation 20 and Equations 23.1-24.2 were transformed into Equation 43 and Equations 44.1-45.2, respectively.

The bunker consumption function demonstrated in Equation 43 is linearized using Algorithm 1. The algorithm was developed as a part of this study solely to linearize the bunker consumption function. The biggest advantage of the algorithm is its capability to control the approximation error. In the initiation phase of the algorithm, u_{max}, u_{min}, ε, and θ are predetermined: u_{max} is the reciprocal of the minimum speed; u_{min} is the reciprocal of the maximum speed; ε is the approximation error, and θ is a very small number required to determine the maximum point of each line segment of the piecewise linear function. In step 13 the slope of the function of the line that connects the minimum and maximum points of the non-linear function was found. Then, to find the point that marks the maximum difference between the non-linear function and the function of the line, the line function is subtracted from the non-linear function and its derivative is taken and equated to zero. The resulting equation can be solved by Newton's method. After the point that maximizes the difference between the non-linear function and the line function was found, the difference is determined. If the difference is bigger than the predetermined approximation error (ε), the maximum point of the nonlinear function is reduced by a small amount (θ) and the whole

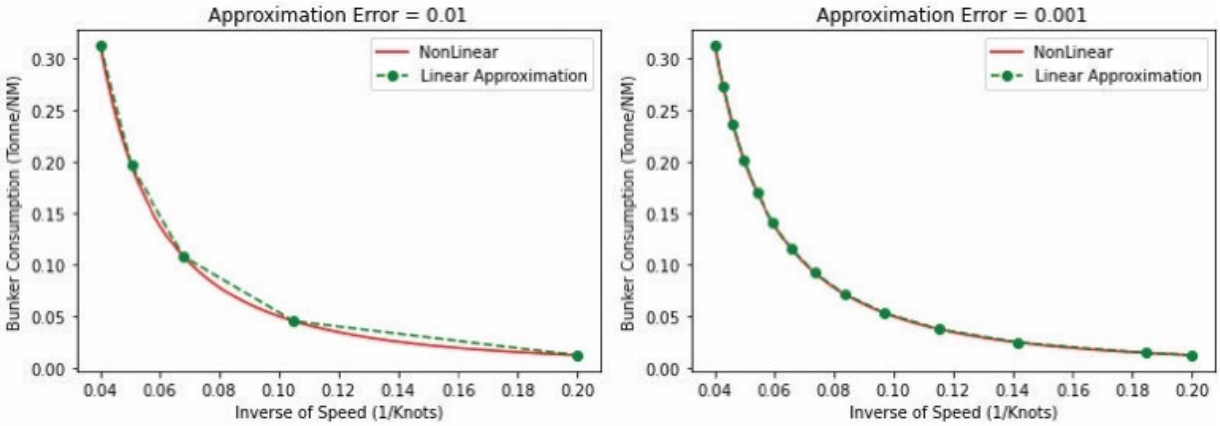


FIGURE 2. Linearization of bunker consumption function.

process is repeated until the difference becomes smaller than the approximation error. When the difference becomes smaller than the approximation error, it means that the maximum point of the first line segment of the piecewise linear function is found. This process is repeated until the maximum point of the last added line segment becomes bigger than the initial maximum point of the nonlinear function.

$$F_{iqv} = \gamma * U_{iqv}^{1-\alpha}/24,\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \tag{43}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,i+1}U_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,i+1}^{str_pass}\delta_{i,i+1}^{str_dur} - (1 - XY_{i,i+1,qv})M,\ \forall i \in P\backslash\{|P|\},\ \forall q \in Q,\ \forall v \in V \tag{44.1}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,minP}U_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,\min P}^{str_pass}\delta_{i,\min P}^{str_dur} - (1 - XY_{i,minP,qv})M,\ i = |P|,\ \forall q \in Q,\ \forall v \in V \tag{44.2}$$

$$T_{ijqv}^{sail_{skip}} \geq dist_{ij}U_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{ij}^{str_pass}\delta_{ij}^{str_dur} - (1 - XY_{ijqv})M - XY_{i,i+1,qv}M,\ \forall i \in P\backslash\{|P|\},\ \forall j \in P, \forall q \in Q,\ \forall v \in V \tag{45.1}$$

$$T_{ijqv}^{sail_{skip}} \geq dist_{ij}U_{iqv} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{ij}^{str_pass}\delta_{ij}^{str_dur} - (1 - XY_{ijqv})M - XY_{i,minP,qv}M,\ i = |P|,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{45.2}$$

After the linearization with the described algorithm, Equations 43-45.2 must be replaced by Equations 46-51.2 to incorporate the piecewise linear approximation into the model. Variable U_{iqv} is replaced with a new variable U_{iqvs} to determine the reciprocal of the speed in each segment of the piecewise linear function. Additionally, other new parameters were added: $slope_s$ are the slope of each line segment; $const_s$ is the constant of each line segment; u_s^{min} is the minimum of the reciprocal of the speed in each segment, u_s^{max} is the maximum of the reciprocal of the speed in each segment; and Y_{iqvs} is the binary variable equals to 1, if the line segment selected, otherwise equals to 0.

$$U_{iqvs} \geq u_s^{min}Y_{iqvs},\ \forall i \in P,\ \forall q \in Q,\ \forall s \in L,\ \forall v \in V \tag{46}$$

$$U_{iqvs} \leq u_s^{max}Y_{iqvs},\ \forall i \in P,\ \forall q \in Q,\ \forall s \in L,\ \forall v \in V \tag{47}$$

$$F_{iqv} = \sum_{s\in L} slope_s U_{iqvs} + \sum_{s\in L} const_s Y_{iqvs}\ (w),\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \tag{48}$$

$$\sum_{s\in L} Y_{iqvs} = 1 - X_{skip_{iqv}},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \tag{49}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,i+1}\sum_{s\in L}U_{iqvs} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,i+1}^{str_pass}\delta_{i,i+1}^{str_dur} - (1 - XY_{i,i+1,qv})M,\ \forall i \in P\backslash\{|P|\},\ \forall q \in Q,\ \forall v \in V \tag{50.1}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,minP}\sum_{s\in L}U_{iqvs} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{i,\min P}^{str_pass}\delta_{i,\min P}^{str_dur} - (1 - XY_{i,minP,qv})M,\ i = |P|,\ \forall q \in Q,\ \forall v \in V \tag{50.2}$$

$$T_{ijqv}^{sail_skip} \geq dist_{ij}\sum_{s\in L}U_{iqvs} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{ij}^{str_pass}\delta_{ij}^{str_dur} - (1 - XY_{ijqv})M - XY_{i,i+1,qv}M,\ \forall i \in P\backslash\{|P|\},\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{51.1}$$

$$T_{ijqv}^{sail_skip} \geq dist_{ij}\sum_{s\in L}U_{iqvs} + \delta_{iqv}^{sea_occ}\xi_{iqv}^{sea_dur} + \delta_{ij}^{str_pass}\delta_{ij}^{str_dur} - (1 - XY_{ijqv})M - XY_{i,minP,qv}M,\ i = |P|,\ \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{51.2}$$

Figure 2 illustrates the nonlinear bunker consumption function and its piecewise approximation. The piecewise linear approximation was conducted using Algorithm 1. In the initiation phase, values of input parameters were predetermined. The parameters are described in Table 3, and they include u_{max}, u_{min}, ε, θ, γ, and α. The values of γ and

TABLE 3. Parameters included in the linearization.

Parameter	Value
γ	0.012
α	3
u_{max}	0.2
u_{min}	0.04
ε	0.01, 0.001
θ	1e-10

α are adopted from the literature [22], [32], [40], [46]. u_{max} and u_{min} are equal to reciprocals of minimum speed and maximum speed. In the linearization process, the minimum speed is considered 5 knots, and the maximum speed is considered 25 knots.

In the linearization, two approximation errors were considered to see the difference between the nonlinear function and its approximation for different error levels. In the first case, ε equals 0.01, and in the second case, ε equals 0.001. An approximation error of 0.001 means that the error for bunker consumption in each leg cannot be more than 0.1% of the fuel cost (fuel price multiplied by leg distance). For example, if the approximation error is 0.001, the leg distance is 200 NM and the fuel price per ton is \$500, the approximation error of fuel costs for that leg cannot be more than \$100 (500∗200∗0.001). θ equals 1e-10. The choice of number for θ does not make a difference if it is a sufficiently small number. Figure 2 illustrates that when ε equals 0.001, the nonlinear bunker consumption function is graphically indistinguishable from its piecewise linear approximation. This shows that the algorithm is working correctly and is very effective.

B. LINEARIZATION OF BILINEAR TERMS

Equations 21 and 38 include bilinear terms (multiplication of two variables) so that they are neither linear nor convex. To be able to solve the model instances these terms must be linearized. They can be linearized very efficiently by using McCormick's inequalities (or McCormick's envelopes) [46], [47]. To include McCormick's inequalities, first, the lower and upper bounds of the variables in the bilinear terms need to be determined. It was assumed that the lower and upper bound for the continuous variables are $F_{iqv} \in \left[l_{F_{iqv}}, u_{F_{iqv}}\right]$ and $H_{iqv} \in \left[l_{H_{iqv}}, u_{H_{iqv}}\right]$. The lower and upper bounds for the binary variables ($XY_{ijqv} \in [0, 1]$ and $H^{rate}_{irqv} \in [0, 1]$) were already defined. McCormick's inequalities for both bilinear terms can be defined as follows:

$$\hat{F}_{ijqv} \geq 0 * F_{iqv} + l_{F_{iqv}} * XY_{ijqv} - 0 * l_{F_{iqv}}, \ \forall i \in P, \ \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (52)$$

$$\hat{F}_{ijqv} \geq 1 * F_{iqv} + u_{F_{iqv}} * XY_{ijqv} - 1 * u_{F_{iqv}}, \ \forall i \in P, \ \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (53)$$

$$\hat{F}_{ijqv} \leq 0 * F_{iqv} + u_{F_{iqv}} * XY_{ijqv} - 0 * u_{F_{iqv}}, \ \forall i \in P, \ \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (54)$$

Algorithm 1 Generation of the Piecewise-Linear Approximation Function

OBJECTIVE
  Find a set of lines to approximate bunker consumption function $Q(u)$
INPUT
  Bunker consumption function, $Q(u) = \gamma u^{1-\alpha}/24, u \in [u_{min}, u_{max}]$
  Derivative of the bunker consumption function, $Q'(u) = \gamma(1-\alpha)u^{-\alpha}/24, u \in [u_{min}, u_{max}]$
  The function of a line that connects two points: $A = (u_1, Q(u_1))$ and $B = (u_2, Q(u_2)) \Longrightarrow G(u) = k(u-u_1) + Q(u_1)$
  Slope $\Longrightarrow k = (Q(u_2) - Q(u_1))/(u_2 - u_1)$
  Intersection $\Longrightarrow c = -ku_1 + Q(u_1)$
OUTPUT
  Set of lines $N = G_1(u), G_2(u), \ldots, G_n(u)$
INITIALIZATION
  $u_{max}, u_{min}, \varepsilon, \theta, \gamma, \alpha$
  $u = u_{max}$
REPEAT
  $k = (Q(u) - Q(u_{min}))/(u - u_{min})$
  $G'(x) - Q'(x) = 0 \Longrightarrow k - \gamma\frac{(1-\alpha)x^{-\alpha}}{24} = 0$ Solve for x
  REPEAT
    $u = u - \theta$
    $k = (Q(u) - Q(u_{min}))/(u - u_{min})$
    $G'(x) - Q'(x) = 0 \Longrightarrow k - \gamma\frac{(1-\alpha)x^{-\alpha}}{24} = 0$ Solve for x
  UNTIL $G(x) - Q(x) < \varepsilon \Longrightarrow k(x - u_{min}) + Q(u_{min}) - \gamma x^{1-\alpha}/24 < \varepsilon$
  RECORD u, k and c ($c \Longrightarrow -k\,u_{min} + Q(u_{min})$)
  $u_{min} = u$
  $u = u_{max}$
UNTIL $u_{min} > u$

$$\hat{F}_{ijqv} \leq 1 * F_{iqv} + l_{F_{iqv}} * XY_{ijqv} - 1 * l_{F_{iqv}}, \ \forall i \in P, \ \forall j \in P, \ \forall q \in Q, \ \forall v \in V \quad (55)$$

$$\hat{H}_{irqv} \geq 0 * H_{iqv} + l_{H_{iqv}} * H^{rate}_{irqv} - 0 * l_{H_{iqv}}, \ \forall i \in P, \ \forall r \in R, \ \forall q \in Q, \ \forall v \in V \quad (56)$$

$$\hat{H}_{irqv} \geq 1 * H_{iqv} + u_{H_{iqv}} * H^{rate}_{irqv} - 1 * u_{H_{iqv}}, \ \forall i \in P, \ \forall r \in R, \ \forall q \in Q, \ \forall v \in V \quad (57)$$

$$\hat{H}_{irqv} \leq 0 * H_{iqv} + u_{H_{iqv}} * H^{rate}_{irqv} - 0 * u_{H_{iqv}}, \ \forall i \in P, \ \forall r \in R, \ \forall q \in Q, \ \forall v \in V \quad (58)$$

$$\hat{H}_{irqv} \leq 1 * H_{iqv} + l_{H_{iqv}} * H^{rate}_{irqv} - 1 * l_{H_{iqv}}, \ \forall i \in P, \ \forall r \in R, \ \forall q \in Q, \ \forall v \in V \quad (59)$$

Equations 21 and 38 must be replaced with Equations 52-59.

C. ROBUST COUNTERPART REFORMULATION

Equations 36 and 50.1-51.2 include uncertain parameters so those equations of the constraints must be reformulated as

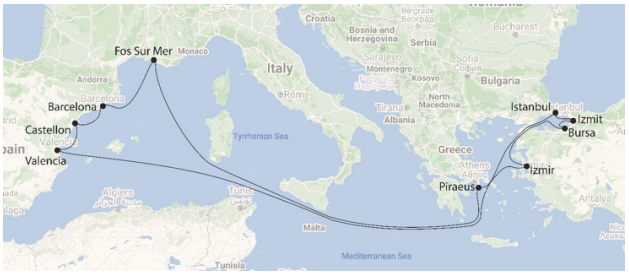


FIGURE 3. Container line service route in the application case.

robust counterparts to solve the model instances. Polyhedral uncertainty set was used as the uncertainty set [48]. The robust counterpart of Equation 36 is demonstrated in Equation 60-61 and the robust counterparts of Equation 50.1-51.2 were demonstrated in Equation 62.1-63.2. For further details of robust counterpart reformulation and their proofs, the reader should refer to [48], [49], and [50]. Equations 36 and 50.1-51.2 were replaced by their robust counterparts i.e., Equations 60-61 and 62.1-62.2.

$$G_{iqv} \geq \tilde{\xi}_{iqv}^{port_{dur}} K_{iqv}^{skip} + W_{iqv} + \Gamma^{port}\theta_{iqv},\ \forall i \in P, \quad \forall q \in Q,\ \forall v \in V \tag{60}$$

$$W_{iqv} + \theta_{iqv} \geq \hat{\xi}_{iqv}^{port_{dur}} K_{iqv}^{skip},\ \forall i \in P,\ \forall q \in Q,\ \forall v \in V \tag{61}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,i+1} \sum_{s \in L} U_{iqvs} + \delta_{iqv}^{sea_occ} \tilde{\xi}_{iqv}^{sea_dur} + \delta_{iqv}^{sea_occ} \Gamma^{sea} \hat{\xi}_{iqv}^{sea_dur} + \delta_{i,i+1}^{str_pass} \delta_{i,i+1}^{str_dur} - (1 - XY_{i,i+1,qv})M,\ \forall i \in P \backslash \{|P|\}, \quad \forall q \in Q,\ \forall v \in V \tag{62.1}$$

$$T_{iqv}^{sail_sec} \geq dist_{i,minP} \sum_{s \in L} U_{iqvs} + \delta_{iqv}^{sea_occ} \tilde{\xi}_{iqv}^{sea_dur} + \delta_{iqv}^{sea_occ} \Gamma^{sea} \hat{\xi}_{iqv}^{sea_dur} + \delta_{i,\min P}^{str_pass} \delta_{i,\min P}^{str_dur} - (1 - XY_{i,minP,qv})M,\ i = |P|,\ \forall q \in Q, \quad \forall v \in V \tag{62.2}$$

$$T_{ijqv}^{sail_{skip}} \geq dist_{ij} \sum_{s \in L} U_{iqvs} + \delta_{iqv}^{sea_occ} \tilde{\xi}_{iqv}^{sea_dur} + \delta_{iqv}^{sea_occ} \Gamma^{sea} \hat{\xi}_{iqv}^{sea_dur} + \delta_{ij}^{str_pass} \delta_{ij}^{str_dur} - (1 - XY_{ijqv})M - XY_{i,i+1,qv}M,\ \forall i \in P \backslash \{|P|\}, \quad \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{63.1}$$

$$T_{ijqv}^{sail_{skip}} \geq dist_{ij} \sum_{s \in L} U_{iqvs} + \delta_{iqv}^{sea_occ} \tilde{\xi}_{iqv}^{sea_dur} + \delta_{iqv}^{sea_occ} \Gamma^{sea} \hat{\xi}_{iqv}^{sea_dur} + \delta_{ij}^{str_pass} \delta_{ij}^{str_pass} - (1 - XY_{ijqv})M - XY_{i,minP,qv}M,\ i = |P|, \quad \forall j \in P,\ \forall q \in Q,\ \forall v \in V \tag{63.2}$$

V. APPLICATION CASE

The model was applied to a container liner service of a leading local container line company. The container line provides services in the Mediterranean, Back Sea, and West Africa regions. Figure 3 illustrates the ports and the route of the container line service. The service includes 9 ports; 4 of which are located in Western Turkey (Istanbul, Izmit, Bursa, and Izmir); 3 of which are located in Spain (Valencia, Castellon, and Barcelona); 1 of which is in France (Fos Sur Mer); and 1 of which located in Greece (Piraeus). The sequence of ports is following the route: Valencia → Castellon → Barcelona → Fos Sur Mer → Piraeus → Istanbul → Izmit → Bursa → Izmir → Valencia. A voyage along the route takes around 21 days.

TABLE 4. Distances between the ports of the service (NM).

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	0	34	161	333	1233	1530	1573	1544	1376
Castellon	34	0	129	302	1224	1521	1564	1535	1367
Barcelona	161	129	0	174	1153	1450	1493	1464	1296
Fos	333	302	174	0	1126	1423	1466	1437	1269
Piraeus	1233	1224	1153	1126	0	336	379	350	185
Istanbul	1530	1521	1450	1423	336	0	49	43	224
Izmit	1573	1564	1493	1466	379	49	0	63	267
Bursa	1544	1535	1464	1437	350	43	63	0	238
Izmir	1376	1367	1296	1269	185	224	267	238	0

The liner service was chosen because it is a typical pendulum liner service route with an average number of ports. On the other hand, the model can be applied to any container liner service with any complexity, short sea or deep sea or with multiple visits to the same port. Even though every port visited only ones in the service, the model can also be applied to services with multiple visits of the same port by assuming each visit to the same port represent a different port. For example, if port of Piraeus would been visited three times in the application case, it could be applied to the model as three distinct ports (such as Piraeus1, Piraeus2, and Piraeus3). Liner shipping companies can apply the proposed model to each of their liner services separately to optimizes all their services.

The proposed model assumes that transshipment of containers can only be done in the ports of the applied service. This assumption is reasonable because inclusion of the ports outside of the liner service will increase the transported distance and fuel consumption. Therefore, they will not be the ones that chosen for transshipment if it will be possible to transship containers at any port in the service.

VI. DATA

In this section parameter data used for instances of the application case that was used for the analysis is described. Table 4 includes distance data for each port pair in the application case. It should be noted that not only distances in the sequence of the service were included but distances between each port pair. This is because non-sequential distances were used in cases where port skipping occurs.

Table 5 includes bunker prices in two cities that are in the region of the container line service route. As of January 01, 2020, the limit on the sulfur content for marine bunkers was reduced from 3.5% m/m to 0.5% m/m by the IMO [51]. After the introduction of this limit, most ships started to use very low sulfur fuel oil (VLSFO) instead of heavy fuel oil (HFO) with ≈3.5% of sulfur content. In Table 6, the fuel price (c^{fuel})

TABLE 5. Values of the parameters.

Parameter	Values	Unit
$\|R\|$	3	N/A
$\|Q\|$	3	N/A
$\|V\|$	8	N/A
c_r^{hand}	75, 60, 45	\$/TEU
c_i^{late}	U[3,000, 5,000]	\$/Day
c^{fuel}	1148.25	\$/t
c_q^{opr}	5,000	\$/Day
cap_q	5,000	TEUs
c_i^{stor}	350	\$/Week
c^{round}	1,500	\$
c^{sea_div}	1,500	\$
e^{fuel}	3.114	N/A
e^{hand}	0.01729	t/TEU
c^{ems}	89.25	\$/t
$\tilde{\xi}_{iqv}^{port_dur}$	0	h
$\hat{\xi}_{iqv}^{port_dur}$	50	h
$\tilde{\xi}_{iqv}^{sea_dur}$	0	h
$\hat{\xi}_{iqv}^{sea_dur}$	25	h
$\xi_{iqv}^{port_dur}$	$\xi_{iqv}^{port_dur} = \tilde{\xi}_{iqv}^{port_dur} + \hat{\xi}_{iqv}^{port_dur}$	N/A
$\xi_{iqv}^{sea_dur}$	$\xi_{iqv}^{sea_dur} = \tilde{\xi}_{iqv}^{sea_dur} + \hat{\xi}_{iqv}^{sea_dur}$	N/A
Γ^{sea}	1	N/A
Γ^{port}	1	N/A
δ_i^{late}	U[5,15]	h
$\delta_{ij}^{str_dur}$	15	h
h_r^{rate}	180, 120, 60	TEU/h
tw^{dur}	8	h
f	7	Day
t_i^{fix}	U[3, 6]	h
$u_{F_{iqv}}$	0.3125	t/NM
$l_{F_{iqv}}$	0	t/NM
$u_{H_{iqv}}$	10,000	TEU
$l_{H_{iqv}}$	0	TEU
M^{late}	72	h
M	504 ($f * \|Q\| * 24$)	N/A

was determined as the average of the two prices shown in Table 5.

Most of the parameter values to be used in the instances of the analysis were illustrated in Table 6. The ones that are not included in the table will be further elaborated on in the following paragraph. The parameters in the table can be explained under five categories: sets, costs, emission factors, and durations. The sets include the number of handling ranges ($|R|$), number of ships ($|Q|$, and number of voyages ($|V|$). The costs include handling costs per TEU for each available handling productivity (c_r^{hand}), cost of late arrival for a ship on each port (c_i^{late}), ship operation cost per day (c_q^{opr}), and container storage cost per TEU at each port (c_i^{stor}). The container handling costs only includes ship to shore or shore to ship transfer costs of terminal gentry crane. This was assumed because only loading and unloading speed affect container ship handing time. The emission factors include the t of CO2 emission generated from t of fuel consumed (e^{fuel}), t of CO2 emissions generated from handling one TEU (e^{hand}), and costs per ton of CO2 (c^{ems}). Those emission parameters equal 3.114, 0.01729 Tran et al., 2017), and \$89.25 (EUR 85.58) [54], respectively. The durations include

TABLE 6. Bunker prices (on 04 July 2022).

City	VLSFO
Istanbul	\$1056
Piraeus	\$1240.5

disruption durations at ports ($\xi_{iqv}^{port_dur} = \tilde{\xi}_{iqv}^{port_dur} + \hat{\xi}_{iqv}^{port_dur}$), disruption durations at sea ($\xi_{iqv}^{sea_dur} = \tilde{\xi}_{iqv}^{sea_dur} + \hat{\xi}_{iqv}^{sea_dur}$), durations resulting from the late arrival of a ship (δ_i^{late}); strait passage durations if available between the ports ($\delta_{ij}^{str_dur}$) container handling rate as the number of containers can be handled per hour (h_r^{rate}); time window durations (tw^{dur}); ship arrival frequency (f); fixed durations at each port for ship berthing (t_i^{fix}). Lower and upper bounds for the variables in the bilinear terms were also determined. Upper and lower bound for the bunker consumption per mile ($l_{F_{iqv}}$ and $u_{F_{iqv}}$) is determined according to the linear approximation parameters found from the application of Algorithm 1 with the parameters described in Table 3. Additionally, upper and lower bounds for container handling amounts ($l_{H_{iqv}}$ and $u_{H_{iqv}}$) were determined according to ship capacities. Because the capacities of each ship are assumed to be 5,000 TEUs, the maximum number of containers that can be unloaded from a ship can be 5,000 and the maximum number of containers that can be loaded at a ship can be 5,000, thus, the maximum number of containers can be handled at a port for a ship's call in the application case is 10,000 TEUs. And the parameters include the maximum allowed late arrival duration for ships (M^{late}) equals 72 hours. The sufficiently big number (M) for modeling purposes was included in Equations 29 and 30 and equals 504. It is found by multiplying f, $|Q|$, and 24.

In addition to the parameters described in Table 6, there were other parameters included and they require more elaboration. Speed parameters i.e., set of line segments for approximating nonlinear fuel consumption function ($|L|$), slopes of those lines ($slope_s$), and constants of those lines ($const_s$). And these speed parameters were generated by applying Algorithm 1 with the parameters described in Table 3 with the approximation error of (ε) 0.001. Additionally, there are parameters for container divergence in case a port is skipped. They include the binary parameter for indicating the availability of diversion between a port pair ($\delta_{ij}^{road_div}$) (The values of the parameter was described in Table 8 in Appendix A); and the parameter for emissions produced from diverted containers ($e_{ij}^{road_div}$). This parameter was calculated for each pair of ports that diversion is available between them: $e_{ij}^{road_div}$ equals $e * dist_{ij}$, where e is the well-to-wheel emission factor for road transport of 1 TEU (Assuming the divergence performed by road and the weight of 1 TEU equals 15 t) and e equals to 0.00135 t of CO2 per TEU-km [54]. $dist_{ij}$ is the distance as kilometers between the ports where the diversion can take place (The values of the distances were described in Table 10 in Appendix A). The other parameters include container transportation demand (d_{ijqv}) between each pair of port calls

TABLE 7. Impacts of collaboration and port skipping.

Instance	Port Skipping	Collaboration		Objective Function			Solution Values			N. of Skipped Ports	N. of Late Arrivals	Avg. Late Time (h)	Avg. Speed (NM/h)
		Handling Rate	Time Windows	Operational Costs (z_1)	Total Costs (z_1+z_2)	Cost of Emissions (z_2)	Emissions (t)	Operational Costs ($)	Total Costs ($)				
1	√	√	√	√			31,915	36,412,349	39,260,780	0	0	0	13.48
2	√	√	√		√		31,378	36,372,006	39,172,529	0	0	0	13.34
3	√	√	√			√	20,576	107,567,338	109,403,722	14	0	0	10.13
4		√	√	√			32,033	36,320,186	39,179,101	-	0	0	13.46
5		√	√		√		31,412	36,370,640	39,174,151	-	0	0	13.32
6		√	√			√	25,584	39,644,040	41,927,440	-	0	0	11.49
7	√		√	√			32,242	36,517,285	39,394,852	1	0	0	13.38
8	√		√		√		32,250	36,520,349	39,398,658	1	0	0	13.41
9	√		√			√	20,575	110,369,192	112,205,520	14	0	0	10.16
10	√	√		√			35,243	39,078,025	42,223,474	2	22	43.19	13.74
11	√	√			√		34,775	39,028,268	42,131,899	2	22	44.09	13.76
12	√	√				√	20,723	104,374,333	106,223,849	14	0	0	10.3
13	√			√			38,764	40,049,943	74,646,643	3	29	44.49	13.98
14	√				√		37,925	40,168,933	43,553,783	4	28	44.41	13.81
15	√					√	20,821	107,635,563	109,493,834	14	2	72	10.38
16		√		√			39,361	39,802,200	43,315,200	-	46	36.76	14.95
17		√			√		39,092	39,816,268	43,305,242	-	44	36.89	14.62
18		√				√	32,994	43,017,607	45,962,367	-	56	72	10.97
19			√	√			32,840	36,908,206	39,839,155	-	0	0	13.47
20			√		√		33,888	36,653,452	39,677,970	-	0	0	13.56
21			√			√	25,616	39,655,873	41,942,137	-	0	0	11.53
22				√			35,499	42,371,044	45,539,296	-	60	34.52	11.85
23					√		35,471	42,402,361	45,568,170	-	67	34.76	12.5
24						√	32,998	43,018,979	45,964,071	-	56	72	11.09

in the container line service excluding the same port pairs; disruption occurrences at ports ($\delta_{iqv}^{port_occ}$); and disruption occurrences at sea ($\delta_{iqv}^{sea_occ}$). The container transportation demand parameter was determined according to the uniform distribution described in Table 11 in Appendix A. Disruption occurrences at ports ($\delta_{iqv}^{port_occ}$) were determined assuming that disruptions occur in 5% of the port calls and disruption occurrences at sea passages ($\delta_{iqv}^{sea_occ}$) determined assuming that disruptions occur 2% of the sea passages. The costs of container diversion per TEU between ports where diversion is available ($c_{ij}^{road_div}$) were described in Table 9 in Appendix A. The parameter for the availability of strait passage ($\delta_{ij}^{str_pass}$) in sea passages between the ports was described in Table 12 in Appendix A.

VII. EXPERIMENTAL FINDINGS

Different instances with different scenarios were built and solved to determine how collaboration and port skipping as a disruption recovery measure affect optimum total costs, operational costs, and emissions. Collaboration of shipping line and terminal operation in the context of this study includes two types: handling rate collaboration and time window collaboration. Handling rate collaboration involves the terminal operator and shipping line agreeing on custom handling rates for each ship, at each port, and on each voyage of those ships as needed. In the context of this study, it was assumed that the terminal operator can provide three handling rates and each handling rate have unit handling costs associated with it, meaning faster handling of containers costs more. The handling rates and their unit handling costs are shown in the previous section in Table 6. Time window collaboration between a shipping line and terminal operator involves the provision of custom time windows in each port, for each ship, and in each voyage of those ships. Both collaboration types, especially time window collaboration requires real-time and constant information sharing regarding actual and estimated durations of disruptions in previous ports and in previous sea passages and estimated and actual port and sea passage times. In the model, handling rate and time window collaboration were modeled including indices for each port, each ship, and each voyage (Tw_{iqv}^{st}, Tw_{iqv}^{st}, and H_{irqv}^{rate}) assuming constant and real-time information sharing between the shipping line and the terminal operators. The instances that did not include collaboration for either handling rate or time window or both included an only index for each port (Tw_{i}^{st}, Tw_{i}^{st}, and H_{ir}^{rate}). This means that optimum handling rate and time windows were only agreed upon between shipping line and terminal operators before the start of the service for each port and they were the same for each ship in each voyage and no real-time information sharing was needed.

The instances in the experimental design shown in Table 7 constructed to systematically reveal the impacts of handling rate collaboration, time window collaboration, and port skipping for any of their availability combination. Considering the availability and unavailability (there are two possibilities for each: available or not) of those three measures, eight possible combinations (2^3) can be constructed. These eight possible combinations evaluated for each objective function, hence, totally twenty-four instances ($8 * 3$) were included in the experimental design.

The model and its instances were built in Pyomo and were solved by Gurobi v9.1.1 by setting the gap value to 0.01. Solutions were performed using Macbook Pro which has M1 CPU with 8 Cores and 16 GB RAM. Most of the instances solved took less than 500 seconds. Only four of the instances described in Table 7 took more than 500 s: Instance 10 took 604 s., Instance 11 took 936 s, Instance 14 took 799 s, and Instance 18 took 1874 s. The solving time of the longest

instance took around half an hour. This shows that different instances of the model can be solved in a reasonable time by a generic solver without needing a customized algorithm. However, applying the model to larger instances may take more runtime for generic solvers and therefore might require a custom algorithm.

A. IMPACT OF COLLABORATION AND PORT SKIPPING

Table 7 describes the results of 24 solved instances. Each instance in the table included a different combination of availability of port skipping, availability of handling rate collaboration, and availability of time window collaboration under three different objective functions: minimization of operational costs ($\min z_1$), minimization of total costs that is the sum of operational and environmental costs ($\min z_1 + z_2$), and minimization of environmental costs ($\min z_2$). And solution values of emission tonnage, operational costs, total costs, number of skipped ports, number of late arrivals, average late arrival durations, and average speed are shown in the table.

Table 7 shows that including environmental costs in the objective function produces a lower tonnage of emissions compared to the minimization of only operational costs. Additionally, total costs in the instances where emissions costs were included in the objective function are lower than the total costs of the instance where only operational costs were included in the objective function. This tells that counting for emissions costs for the optimization of container shipping scheduling reduces emission tonnages as well as total costs because shipping lines must bear emission costs even if they have not considered it while optimizing their ship schedules. The inclusion of emission costs can help to reduce emission tonnage, but it is limited and far away from minimum emissions that can be achieved by minimizing only emission costs. Table 7 illustrates that the minimum achievable emission tonnage is 20,575 t. It is achieved only in Instances 3 and 9. However, this minimum emission tonnage can be achieved by increasing operational costs by 195.6% by almost tripling it. The increase in the operational cost is bigger for Instance 9 because it excludes handling rate collaboration. In both instances, the reduction in the emission tonnage was achieved by skipping many ports (14) and in this way enabling speed reduction to around 10 NM. The table also shows that in general when port skipping is available, the minimization of emissions results in a high number of ports skipping to reduce ship speed. This indicates that minimization of emissions tonnage not only triples the operational costs but also hurt customer service.

The results showed that port skipping as a disruption recovery is mostly effective for emission reduction. The comparison of Instances 13-14 with Instances 22-23 shows that port skipping is effective for disruption recovery in terms of operational costs when there is no collaboration taking place between the shipping line and the terminal operators. It reduced operational costs by 5.5%, and total costs by 5.3%. The comparison of Instances 1-2 with Instances 4-5

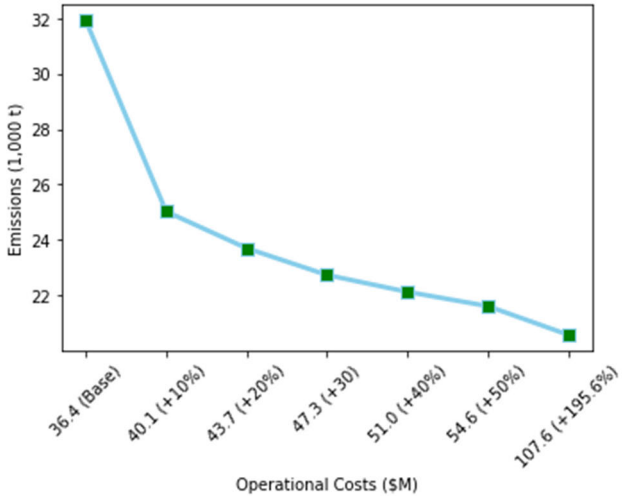


FIGURE 4. Trading off operational costs for reducing emission.

suggests that when collaboration takes place, port skipping had almost no impact. On the other hand, when a similar comparison is made with the instances where only the emissions were minimized as in comparing Instance 15 with 25, the availability of port skipping decreased emissions by 37% while increasing the operational costs by 150% and total costs by 138%. When the same comparison is made with Instances 3 and 6, it shows that port skipping created less reduction in emissions only 19.6% while creating a higher increase in operational costs and total costs by 171% and 161%, respectively. This suggests that when the shipping line and terminal operators share real-time information and collaborate, there would be a rare need for port skipping except when only the costs of emission were minimized. The comparison of Instances 13-14 with Instances 22-23 demonstrates that when there is no collaboration taking place, the availability of port skipping reduced late arrivals of ships. Containers for the skipped ports arrived late, but the number of late arrivals in other ports was considerably reduced. This might increase customer service in general.

In Table 7, the comparison of Instances 16-17 with Instances 22-23 shows that collaboration for only handling rate increased emissions tonnage but reduced operational costs and total costs. The comparison of Instance 18 with Instance 24 suggests that handling rate collaboration has no impact when the emissions are minimized compared to no collaboration alternative. Additionally, handling rate collaboration compared to no collaboration reduced the number of late arrivals, however, the duration of late arrivals was increased. The comparison of Instance 3 with Instance 9 suggests that minimum emissions can be achieved by 2.5% less total costs when there was handling rate collaboration taking place.

Comparison of Instances 19-20 with Instances 22-23 suggests that time window collaboration can reduce both emission tonnage and total costs by 7.5% and 13%, respectively, compared to non-collaboration and non-availability of port

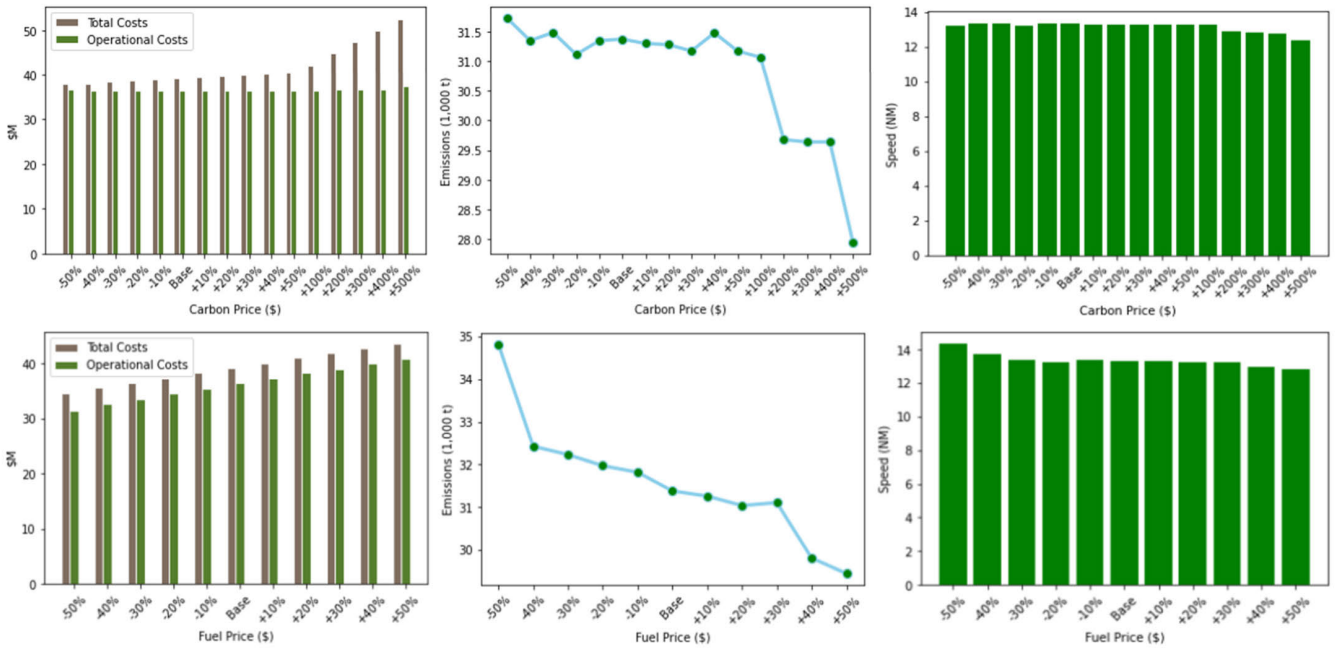


FIGURE 5. Sensitivity of total costs, emissions, and average speed to carbon and fuel prices.

skipping alternatives. The comparison of Instance 21 with Instance 24 suggests that when only emissions were minimized, time window collaboration provided the best solution again by reducing emissions by 22.4% while also reducing total costs by 8.8%. Additionally, comparing Instances 1-3 with the instances in the whole table demonstrates that combining time window collaboration with handling rate collaboration provided the best results in terms of both operational costs and emission tonnage. Time window collaboration also eliminated late arrivals; however, the ships can still be late according to the schedule provided to the shippers of the containers. These results suggest that customer service can be improved by real-time information sharing with shippers because the disruptions occurrences are not under the control of container shipping lines and can happen in services of any container shipping line.

The results in the table show that the most effective disruption recovery option in terms of both operational costs and emission tonnages was the time window collaboration. Availability of port skipping slightly reduced emission tonnage while having more impact on operational costs. And port skipping was not required when collaboration takes place between the shipping line and terminal operators. On the other hand, handling rate collaboration had only a positive impact on operational costs while slightly increasing emission tonnage when considered without port skipping or time window collaboration.

B. TRADING OFF OPERATIONAL COSTS FOR REDUCING EMISSION

Table 7 shows the minimization of emission costs caused a tremendous increase in operational costs by tripling them. On the other hand, the minimization of operational costs created emissions that are 60% higher than the minimum emissions resulting from the minimization of emission costs. This section evaluated that how can bearing certain amounts of operational cost impact emission tonnage reduction. For that purpose, the lexicographic goal programming approach was used [55]. In lexicographic goal programming, there is a number of priority levels. Objective function of each priority level is optimized, and the optimum value is added as a constraint. First, operational costs were minimized then it was added to the model instance as a constraint as in Equation 64 that the operational costs are smaller or equal to the minimum operational costs (z_1^*). The minimum operational cost (z_1^*) was found by the minimization of operational costs. Then, the emission costs were minimized to find the minimum tonnage of emissions that can be achieved with the minimum operational costs. This was repeated by gradually increasing minimum operational costs (z_1^*) to find how much emissions can be reduced by bearing a certain amount of operational cost increases. Figure 4 illustrates the result of this cost trade-off analysis.

$$z_1 \leq z_1^* \tag{64}$$

Figure 4 illustrates that with the minimum operational costs, which equals $36,412,349, emissions can be minimized at most down to 31,915 t. On the other hand, by bearing only a 10% increase in the operation costs, the emissions can be reduced to 25,021 t which is a 21.6% reduction. However, additional sacrifices in operational costs can bring gradually lesser reductions in emissions. For example, bearing another 10% operational cost can reduce emissions to 23,698 t which is a 25.7% reduction. To reduce emissions to their minimum

achievable level, operational costs must be increased by 195.6%, almost tripling it.

This analysis demonstrates that it is possible to achieve a big portion of emission reduction by small sacrifices in operational costs. However, reducing the emissions to their minimum level only with operational measures without adopting technological improvements would require tremendous increases in operational costs.

C. SENSITIVITY OF TOTAL COSTS, EMISSIONS, AND AVERAGE SPEED TO CARBON AND FUEL PRICES

Figure 5 illustrates the sensitivity of total costs and emissions tonnage to carbon prices. Carbon prices are applied in emission trading schemes to reduce industrial emissions. It is thought that increasing carbon prices can provoke companies to reduce and eventually eliminate their emissions. In this section, the effectiveness of increasing carbon price on emission was evaluated with its impact on total costs, operational costs, and average speed. For this reason, the total costs were minimized by including varying carbon prices. The first graph in the figure shows that increasing carbon prices almost had no impact on operational costs but increased total costs. This is understandable because the total costs include operational costs and emission costs, an increase in the price of carbon increases emission costs thus the total costs. However, when both graphs are investigated together, they provide more information. For the increase in the carbon price to be effective in emission reduction, it must be increased by at least 200%. Even then, the emission reduction was 4.5%. Another decrease in the emission tonnage by 5.6% is possible when the carbon price was increased by 500%. The third graph in the figure confirms the relationship between speed and emission production. It is also shown in the figure that to trigger a meaningful reduction in vessel speed, carbon prices need to be increased 200%. This emission reduction through increasing carbon prices can tremendously increase total costs. Additionally, decreases in carbon prices had a slight impact on emission tonnage.

These results indicate that for a carbon price increase be effective in emission reduction, the increase must be at least 200%. Even then, it has a limited impact on encouraging emission reduction through operational measures. However, an increase in carbon price increased the minimum total costs, and this can encourage shipping lines to adopt technological innovations for emission reduction. On the other hand, a decrease in carbon price can make shipping lines abandon operational practices for emission reduction.

Figure 5 also evaluates the sensitivity of total cost, emissions, and average speed to fuel prices. The fourth graph in the figure shows that fuel price increases or decreases had a big impact on both total and operational costs. On the other hand, emissions were more sensitive to fuel price decreases compared to fuel price increases. Decreases in fuel prices down to 50% could create a big increase in emissions by 11%. And 6.2% decrease in emissions can be possible when fuel price was increased by 50%. The last graph in the figure shows that average speed is more sensitive to fuel price compared to carbon price. This is because emission production highly depends on vessel speed. These findings indicate that when the fuel price decreases, it is more profitable for container lines to abandon operational measures that reduce emissions such as slow steaming.

VIII. DISCUSSION

This study aimed to propose a comprehensive green container shipping scheduling model that accounts for uncertainties in port and sea passage times, disruptions, and collaboration of carrier and terminal operators. The experimental results demonstrated that the most effective disruption recovery measure in terms of both operational costs and emission tonnages is the time window collaboration. It can reduce both emission tonnage by 22.4% and total costs by 8.7%. These results are consistent with the findings of previous studies that demonstrated higher availability of port time windows reduces operational costs [19]. Handling rate collaboration reduces total costs while increasing emission tonnage when considered without port skipping or time window collaboration. A similar result was found by Liu et al. [35] who suggested collaborating on the provision of handling rate considerably reduces costs. The advantages of both handling rate and time window collaboration are also underlined in other studies [15]. Our findings also suggest that the introduction of port skipping without collaboration reduces total costs by 4.4%, while emissions are minimized, skipping reduces emission tonnage by 37%. The effectiveness of port skipping for disruption recovery in terms of costs was also observed by Li et al. [29]. However, our findings demonstrated that port skipping has almost no impact when collaboration takes place between the shipping line and terminal operators. This is because there would be a rare need for port skipping when a shipping line and terminal operators share real-time information and collaborate on the provision of custom time windows and handling rates in each port, for each ship in each voyage.

Additionally, the lexicographic goal programming approach was used to reveal the amount of emissions reduction that could be achieved by trading off operational costs. Its results demonstrated that bearing a 10% increase to the minimum operational cost can reduce emissions by 21.6%. However, operational costs must be increased by 195.6% to reduce emissions to their minimum achievable level. These results indicate that a big portion of emission reduction can be achieved by small sacrifices in operational costs. On the other hand, reducing emissions to their minimum achievable level can create tremendous operational cost increase. This implies that reducing emissions to their minimum level requires adopting technological improvements not merely operational measures. Further evaluations suggested that total

costs and emissions are more sensitive to fuel price changes than changes in carbon price. Similar outcomes were found by Dulebenets [15]. Although, their analysis only considered operational costs. Findings related to carbon prices suggest that an increase in carbon prices must be 200% to trigger a 4.5% reduction in emissions. This suggests that even a 200% increase in carbon prices can have a very limited impact. These findings contradict the findings of Dulebenets [16] as their study suggested that carbon prices can significantly affect the design of vessel schedules. The reason behind that contradiction is the fact that the sensitivity analysis in this study used the instance that considers both handling rate and time window collaborations, but their analysis had not accounted for any collaboration. By collaborating with terminal operators in terms of handling rate and time window provision, the container shipping line can already take advantage of slow steaming. And, understandably, further exploitation of slow steaming would require a big increase in carbon prices.

This study can be further improved in various ways. First, demand for container transportation can vary seasonally, and economic crises or global health crises such as Covid-19 can have overwhelming impacts on it. Demand changes can have a big impact on container line service schedules; therefore, the stochasticity of the demand parameter can be included in the model from a stochastic constraint programming perspective to investigate that impact. Second, the findings of this study demonstrated that reducing emissions can tremendously increase operational costs, therefore, further studies can investigate a sharing mechanism between container lines and shippers for the increased operational costs by evaluating how much more shippers are willing to pay for a container transportation service conducted by producing lower emission in certain degrees. Third, the application of the model to a larger application case might increase the runtime of the generic solver, therefore, further studies can provide a customized algorithm for efficiently solving larger instances. Fourth, the model designed for individual container liner companies to monitor and optimize their own carbon emissions as a part of their corporate social responsibility independent of a carbon levy. However, future studies can model specific implementation of a carbon levy.

APPENDIX A
INPUT DATA

TABLE 8. Availability of diversion between the ports of the service.

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	0	1	1	1	0	0	0	0	0
Castellon	1	0	1	1	0	0	0	0	0
Barcelona	1	1	0	1	0	0	0	0	0
Fos	1	1	1	0	0	0	0	0	0
Piraeus	0	0	0	0	0	0	0	0	0
Istanbul	0	0	0	0	0	0	1	1	1
Izmit	0	0	0	0	0	1	0	1	1
Bursa	0	0	0	0	0	1	1	0	1
Izmir	0	0	0	0	0	1	1	1	0

TABLE 9. Costs of diversion between the ports of the service.

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	N/A	200	300	600	N/A	N/A	N/A	N/A	N/A
Castellon	200	N/A	200	300	N/A	N/A	N/A	N/A	N/A
Barcelona	300	200	N/A	300	N/A	N/A	N/A	N/A	N/A
Fos	600	300	300	N/A	N/A	N/A	N/A	N/A	N/A
Piraeus	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Istanbul	N/A	N/A	N/A	N/A	N/A	N/A	100	200	500
Izmit	N/A	N/A	N/A	N/A	N/A	100	N/A	200	500
Bursa	N/A	N/A	N/A	N/A	N/A	200	200	N/A	500
Izmir	N/A	N/A	N/A	N/A	N/A	500	500	500	N/A

TABLE 10. Road distanced between the ports in between diversion is possible.

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	N/A	82	464	800	N/A	N/A	N/A	N/A	N/A
Castellon	82	N/A	330	726	N/A	N/A	N/A	N/A	N/A
Barcelona	464	330	N/A	460	N/A	N/A	N/A	N/A	N/A
Fos	800	726	460	N/A	N/A	N/A	N/A	N/A	N/A
Piraeus	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Istanbul	N/A	N/A	N/A	N/A	N/A	N/A	120	161	486
Izmit	N/A	N/A	N/A	N/A	N/A	120	N/A	104	407
Bursa	N/A	N/A	N/A	N/A	N/A	161	104	N/A	372
Izmir	N/A	N/A	N/A	N/A	N/A	486	407	372	N/A

TABLE 11. Transportation demand between the ports of the service.

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	-	U[0,25]	U[0,25]	U[0,25]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]
Castellon	0	-	U[0, 25]	U[0,25]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]
Barcelona	0	0	-	U[0,25]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]
Fos	0	0	0	-	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]
Piraeus	U[100,250]	U[100,250]	U[100,250]	U[100,250]	-	U[100,250]	U[100,250]	U[100,250]	U[100,250]
Istanbul	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	-	U[0,25]	U[0,25]	U[0,25]
Izmit	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	0	-	U[0, 25]	U[0,25]
Bursa	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	0	0	-	U[0,25]
Izmir	U[100,250]	U[100,250]	U[100,250]	U[100,250]	U[100,250]	0	0	0	-

TABLE 12. Availability of strait passage between the ports of the service.

	Valencia	Castellon	Barcelona	Fos	Piraeus	Istanbul	Izmit	Bursa	Izmir
Valencia	0	0	0	0	0	1	1	1	0
Castellon	0	0	0	0	0	1	1	1	0
Barcelona	0	0	0	0	0	1	1	1	0
Fos	0	0	0	0	0	1	1	1	0
Piraeus	0	0	0	0	0	1	1	1	0
Istanbul	1	1	1	1	1	0	0	0	1
Izmit	1	1	1	1	1	0	0	0	1
Bursa	1	1	1	1	1	0	0	0	1
Izmir	0	0	0	0	0	1	1	1	0

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